Department of Education

STEM Education
National Schools of Excellence
Students' Resource Book
Grades 12
Term 1

DOMAIN	PHYLA
Science	Physics
Technology	
Engineering	
Mathematics	

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THEMA: BIOTECHNOLOGY

Introduction

Biotechnology is a branch of biology and chemistry that employs mathematics and physics to develop tools and engineering insights for modern biology and biomedical research. It entails the study and application of living cells and cellular materials in the development of pharmacological, diagnostic, agricultural, environmental, and other products that promote human welfare and society. It's also utilized to examine and manipulate genetic information in animals in order to imitate and study human disease. Biotechnology is a science that is both fundamental and applied. Agriculture, health care and medical sciences, biomedical engineering, cell biology, immunology, seed technology, virology, plant physiology, forensics, industrial processing, and environmental management all benefit from the technologies it develops.

There are a number of career paths linked to the field of Biotechnology, such as: working in research labs, graduate training in biology and chemistry, and professional careers in the health sciences. If you are someone who enjoys STEM (particularly biology and chemistry), below are some best biotechnology careers that may be of interest to you (research on them for more information):

- Biomedical Engineer
- Biochemistry
- Medical Scientist
- Clinical Technician
- Microbiologist
- Process Development Scientist
- Biomanufacturing Specialist
- Business Development Manager
- Product Strategist
- Biopharma Sales Representative
- Medical Scientist
- Biotechnological Technician
- Epidemiologist
- Microbiologist
- Medical and Clinical Lab Technologist
- Biomanufacturing Specialist
- Bioproduction Specialist

MODULE 12.1: BIOPHYSICS

4-5 Weeks

Biophysics is a vibrant scientific field of study that comprises a variety of disciplines including mathematics, chemistry, engineering, material science and other life sciences. The primary aim of this field is to utilize and apply the principles and laws of physics to understand how biological systems work.

In this module, the basic concepts of biophysics will be covered that describes how this field acts as a bridge between the science domains (i.e., Physics, Biology and Chemistry). This will establish a clear understanding to then discuss its influence as a budding field of science, and its significant contributions to proteomics (protein science) and genomics (genetics) which have broader applications in biotechnology.

Prerequisites:

- **Grade 11 Term 2, STEM Engineering (Thema Biotechnology)**

Bridging Phyla(s):

- **Grade 12 Term 1, STEM Biology (Thema Biotechnology)**
- **Grade 12 Term 1, STEM Chemistry (Thema Biotechnology)**

Topic 1: Introduction to Biophysics

In this topic, an overview of the field of biophysics will be introduced and several key applications will be highlighted. In addition, genomics and proteomics will also be introduced as a major sub-field with key applications in biophysics that will later be discussed in greater detail in the subsequent topics (Topics 2 and 3) of this module.

Key Points:

- In biophysics, the theories and methods of physics are applied to understand how biological systems work.[1]
- Proteomics and Genomics important are complementary sub-fields in molecular biology and are notable applications of biophysics.

1.1. Introduction

1.1.1. *Definition*

Biophysics is a scientific field that describes how physical forces and changes in the mechanical properties of cells, tissues, and their surrounding microenvironment can contribute to cell development; how cells rapidly reproduce (proliferation), and reach maturity in form and function (differentiation).[2]

1.1.2. **Biophysics: The Bridging Science**

Biophysics integrates a variety of broad disciplines including Biology, Chemistry, Computer Science and Engineering. In this manner, biophysics establishes a bridge that connects and applies the theories and methods of Physics in each discipline to understand how biological systems work. Biophysics has been critical to understanding the mechanics of how the molecules of life are made, how different parts of a cell move and function, and how complex systems in our bodies, such as the brain, circulatory system, immune system, as well as the larger ecosystem work.

Currently, biological research involves a meticulous ordeal of experiments that produce large amounts of data. This can prove to be very tedious for biologists to understand, interpret and predict from the data on how these systems might work.

As such, biophysicist develop a broad range of tools that are capable of performing these tasks. They are uniquely trained in the quantitative sciences of the disciplines mentioned and they are able tackle a wide array of topics, ranging from how nerve cells communicate, to how plant cells capture light and transform it into energy, to how changes in the DNA of healthy cells can trigger their transformation into cancer cells, and many other biological problems.

1.2. Applications of Biophysics

[1] *What Is Biophysics? | The Biophysical Society*

[2] *(Biophysics and Biomechanics, n.d.). https://www.bruker.com/it/applications/academia-life-science/cell-biology/biophysics-and-biomechanics.html*

Biophysicists work to utilize updated technologies to design and develop methods to solve a variety of scientific mysteries and solve many of our global challenges. This can include the prevention and treatment of certain diseases, creating genetically modified crops to eradicate global hunger and produce renewable energy sources for sustainability. In short, biophysicists are at the forefront of solving age-old human problems as well as problems of the future.

Listed below are some significant contributions of biophysics that have assisted in understanding biological systems.

1. **DNA Structure**

The structure of the Deoxyribonucleic Acid (DNA) was solved in 1953 using biophysics, and this discovery was critical to showing how DNA is the molecule that carries genetic information from one generation to another and is thus the blueprint for life.

Since its discovery, we are now able to read the sequences of DNA from thousands of humans and all varieties of living organisms. To analyze and manage the enormous quantities of data generated from this sequence, biophysical techniques are used.

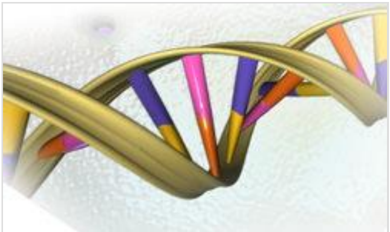

Fig. SP-BT-TRB-1.1: Strand of DNA (Source: What Is Biophysics? | The Biophysical Society)

Note
• *Protein and Gene sequencing will be discussed more in Topic 1.2: Proteomics and Genomics.* • *Additionally, Topic 2 and 3 will further expand and highlight the significance and methods used in these scientific studies.*

2. **Computer Modelling**

Biophysicists develop and use computer modelling methods to visualize and manipulate the shapes and structures of proteins, viruses, and other complex molecules. The information obtained are vital and necessary to develop new drug targets.

An example of this application can be performed on patients with tumors: understanding how proteins mutate and cause them to grow.

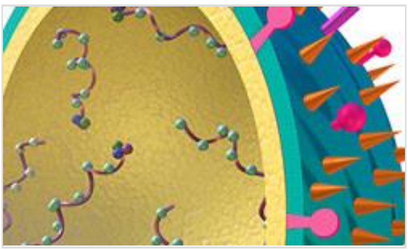

Fig. SP-BT-TRB-1.2: Computer model of a protein structure. (Source: What Is Biophysics? | The Biophysical Society)

3. Molecules in Motion

Biophysicists study how hormones move around the cell, and how cells communicate with each other. Using fluorescent tags, biophysicists have been able to make cells glow like a firefly under a microscope and learn about the cell's sophisticated internal transit system.

4. Neuroscience

Biophysicists are building computer models called neural networks to model how the brain and nervous system work, leading to new understandings of how visual and auditory information is processed. A bulk of the development is actually contributed from Computer Science and Mathematics that aimed to replicate this form of intelligence onto digital systems called Artificial Intelligence. Based on these models, biophysicists incorporate data

Note

- *This will be covered in greater detail in THEMA: ARTIFICIAL INTELLIGENCE.*

5. Medical Applications

a. Biomechanics

Biophysics has also been critical to understanding biomechanics such as the movement and contraction of muscles, tendon-ligament and joints (bones) connection, and their physical properties (e.g., viscoelasticity and strength). This information is useful in physiotherapy, particularly in the design of better prosthetic limbs.

b. Nuclear Medicine, Imaging, Diagnostics and Therapy

Biophysics is at the forefront of research that is transforming our understanding of biology and the practice of medicine in brilliant ways. Biophysicists have developed sophisticated diagnostic imaging techniques including MRIs, CT scans, and PET scans. Biophysics continues to be essential to the development of even safer, faster, and more precise technology to improve medical imaging and teach us more about the body's inner workings.

Fig. SP-BT-TRB-1.3: Computer model of a protein structure. (Source: What Is Biophysics? | The Biophysical Society)

Biophysics has been essential to the development of many life-saving treatments and devices including kidney dialysis, radiation therapy, cardiac defibrillators, pacemakers, and artificial heart valves.

Note

- *Nuclear Medicine will be covered in greater detail in THEMA: HEALTH AND MEDICINE.*
- *The application of Cardiac Defibrillator and pacemaker will be covered in the next MODULE: Bioelectronics.*

6. Genomics and Proteomics

Genomics and Proteomics is a particularly useful study with key applications of biophysics. They are two major sub-fields that share similar concepts and applications in the field of molecular biology; they are essentially complementary. These sub-fields are vital in understanding the function and structure of any living organism.

Fundamentally, genomics involves the scientific study of genomes which are the biological materials from which an organism is made up of. Genomes contain genes that store the information (genetic code) about the DNA structure (genetics) of organisms. The research conducted to uncover the information contained within the genome is referred to as genomics.

Based on the information stored within the genome, the DNA is translated into mRNA that specifies the nucleotide sequence. Subsequently, this sequence determines the sequence of an amino acid which creates proteins. The entire set of expressed proteins in an organism is called a proteome. The research conducted to determine the specific characteristics, structure, functions and expressions of the complete proteins in the cell is known as proteomics.

Difference between Genomics and Proteomics – Genomics vs Proteomics

The main difference between these two sub-fields is that genomics examines the genes of an organism, whereas proteomics analyses the whole set of proteins within cells. Genomic studies are essential to comprehend the function, structure and location of genes in an organism. Proteomics studies are useful because proteins are the true functional molecules of cells, and they represent the real-time physiological conditions.

Table SP-BT-TRB-1.1 highlights the main differences between these two sub-fields.

Table SP-BT-TRB-1.1: Main Differences between Genomics and Proteomics

Character	Genomics	Proteomics
Definition	Genomics is the scientific study of genomes, which refers to the genetic material that is present within a living organism or cell.	Proteomics is the scientific study that involves that examination of the proteins that are expressed by the genome of an organism.

Research into	The study of genes within an organism.	The study of all proteins that make up a cell.
Unit under Study	Research into the functions of genomes	Study of the functions of proteomes.
Nature of Study Material	The genome is always constant. Every single cell in an organism is able to have the identical gene set.	The proteome is ever-changing and fluctuates. The number of proteins made in various tissues differs depending on the expression of genes.
Utilization of High Throughput techniques	High-throughput technologies are utilized in genomics field to create maps, sequences, and analyses of genomes.	In proteomics, the characterization of the 3D structure as well as the role of proteins is performed by means of high-throughput methods.
Techniques involved	The methods used in genomics encompass strategies for gene sequencing including targeted gene sequencing and whole genome shotgun sequence as well as the construction of express sequence tags (ESTs) and the identification SNPs, or single nucleotide polymorphisms (SNPs) and an analysis of and interpretation of the sequenced information using various databases and software.	Techniques involved include the extraction and electrophoretic extraction of proteins. The digestion of proteins using trypsin to break them into smaller fragments as well as the determination of amino acid sequence through mass spectrometry, and the identification of proteins based on the information contained in the protein databases. Furthermore, the 3D shape of the protein could be predicted by using software-based techniques. Protein expression can be studied using microarrays of protein. Protein-network maps can be created to study interactions between proteins and proteins.
Types	Two types of Genomics: • Structural Genomics • Functional Genomics.	Three types of Proteomics: • Structural • Functional • Expression
Important Areas	Genome sequencing projects like The Human Genome Project are among the most important fields of genomics.	Proteome database development, such as SWISS-2DPAGE, as well as the development of software for computer-aided drug design are most important fields of proteomics.

In the subsequent topics (Topic 2 and 3), we will cover these two concepts in detail.

Activity 1: Exercise

Answer the following questions:

Questions

1. **Define the following terms:**
   i. **Biophysics**

   ii. **Genomics**

   iii. **Proteomics**

2. **State and briefly explain at least 2 applications that employ Physics laws and theories in biological systems.**

3. **What is the difference between a gene and a genome?**

4. **Why are Genomics and Proteomics described as complementary sub-fields of Molecular Biology?**

Activity 2: Class Discussion

In this activity, place the students into two groups to discuss about topic of DNA and Protein Sequencing. Allow the students to formulate discussions as they try to discover the benefits and drawbacks in using DNA and Protein Sequencing.

The following questions below can serve to guide the discussion.

1. **What are the benefits of Gene and Protein Sequencing?**
2. **What are the drawbacks of Gene and Protein Sequencing?**
3. **Why are we not able to clone another human being?**

Topic 2: Genomics – DNA Sequencing

In this topic, the concepts of DNA sequencing will be covered. The objective is to relate the concepts of DNA sequencing taught in Biology and the different techniques employed in Physics that microbiologists use to identify the sequence of DNAs. This will illuminate students to develop a broader understanding of the applications of physics in this field.

Key Points:

- Genomics is the study of genes, which is the biological materials that store information that are necessary for creating life.
- DNA sequencing refers to the analysis and procedures involved in deciphering the sequence of the nucleotides within a gene.
- In sanger sequencing, the DNA of interest is replicated many times, producing fragments of varying lengths. Dye colored or fluorescent "chain terminating" nucleotides indicate the end of these fragments and allow the DNA sequence to be determined.

2.1. Introduction

Fundamentally, genomics is the scientific study of genes, which is a short strand or region of DNA. DNA is the primary unit of heredity and provides the genetic information necessary for creating organisms of all types. The basic building blocks of DNA are known as nucleotides and a sequence of these nucleotides is what forms a gene. Thus, a gene is a distinct stretch of DNA that encodes the synthesis of functional molecules (amino acids or proteins).

Genomics involves the use of recombinant DNA technology, DNA sequencing, and Bioinformatics to study the nature and function of the complete set of genes or genome. The studies conducted on genes are crucial in understanding most of the human biological systems and problems. This can include the investigation and diagnosis of genetic disorders and mutations, before or after conception. Geneticist can find the genes responsible for the diseases and work towards developing a therapeutic drug or possible cure. Genomics (using DNA sequencing) can also be used in forensics to help scientists investigate crimes using DNA samples (e.g., hair, blood, fingerprints, etc.) that were left on the crime scene.

Moreover, applications of gene sequencing can have a wide range of applications in agriculture and industrial biotechnology.

Therefore, studies on genomics are of great importance. We will focus our study of genes on the basics of DNA Sequencing, and relate the applications of physics involved in this method.

Note

- *This topic will only highlight the concepts of DNA sequencing. The basics should be covered in greater detail in Gr 12. Biology (THEMA: BIOTECHNOLOGY - MODULE: 12.1 - Bioinformatics). Refer students to their notes on this.*

2.2. DNA and DNA Sequencing

DNA Sequencing is a method of determining the order in which the nucleotides are arranged along a strand of DNA. DNA has a double helical structure that is composed of four nucleotide bases that always exist in the complementary pairs. The nucleotide adenine (A) always pairs with thymine (T), and the nucleotide cytosine (C) always pairs with guanine (G) *(see Fig.SP-BT-TRB-2.1)*. These pairs form the basis of DNA molecules and the order, or sequence, of these pairs tells the cells how to behave. This concept lies at the core of the DNA sequencing methods.

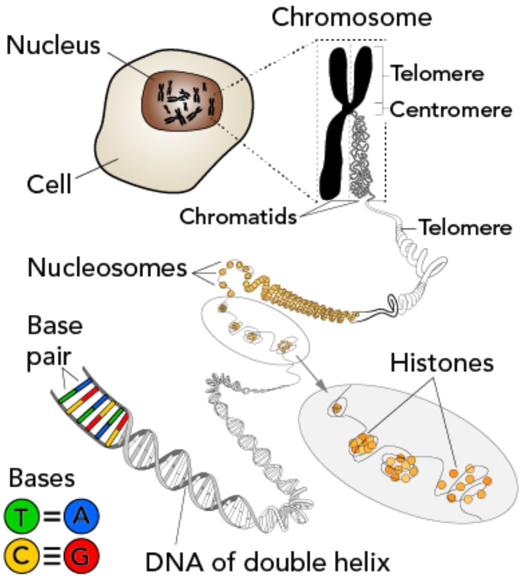

Fig. SP-BT-TRB-2.1: Graphic decomposition of a chromosome to the base pair of the DNA (Source: Let's Talk Science, adapted from illustration by KES47 via Wikimedia Commons).

2.2.1. Brief History of DNA Sequencing

In the late 1970s, scientists developed the first two methods of DNA sequencing. The first method is called Maxam-Gilbert sequencing and was named after its developers, Allan Maxam and Walter Gilbert. Their method involved the use of chemicals to break DNA into small chunks to determine its sequence. This method was revolutionary at the time, but is inefficient in comparison to the newer methods that developed thereafter.

In the same decade (1977), Frederick Sanger and his colleagues developed another method for sequencing DNA. Their method was the most widely used for about 40 years.

Sanger sequencing enabled scientists to read the genetic code for the first time and because of this ground-breaking discovery, Frederick Sanger was awarded a Nobel Prize in Chemistry. The sanger sequencing is based on the process of DNA replication. Scientists make copies of DNA strands and then observe which nucleotides are added. In this way, the nucleotide sequence can be seen.

Sanger sequencing is a well-established method of DNA sequencing that was heavily utilized in the human genome project. This project took about 14 years (1990 – 2003) to complete and involved a united front of international researchers with the objective to completely map all of the genes, and thus complete the genome of the human species (homo sapiens).[3]

2.2.2. Sanger Sequencing

Although there are faster and cheaper methods available today, the sanger sequencing method is still a popular method that forms the basis for modern computer automated sequencing techniques. As such, this topic will focus on explaining this method.

2.2.2.1. <u>Ingredients</u>

Sanger sequencing involves replicating DNA or creating identical copies of a particular gene. The basic ingredients used include a:

- **Template Strand**
    - This is a single-stranded DNA (ssDNA) of interest to be sequenced. It is usually obtained from DNA amplification.
- **DNA Polymerase**
    - this is an enzyme that is responsible to catalyze the synthesis of replicating DNA.
- **DNA Primer**
    - It is a short ssDNA that binds onto the template strand to indicate the initial point to begin sequencing.
- **Deoxy Nucleotides**
    - They are referred to as Deoxynucleotide triphosphate (dNTPs) that involve the DNA base pairs. In DNA sequencing, four dNTPs are used:
        - Deoxyadenosine triphosphate (dATP)
        - Deoxythymidine triphosphate (dTTP)
        - Deoxycytidine triphosphate (dCTP)
        - Deoxyguanosine triphosphate (dGTP)
- **Dideoxy Nucleotides**
    - They are the modified versions of dNTP and are labelled as dideoxy-NTP (ddNTP). They are used to terminate the long DNA chain synthesis of all the four nucleotides. They include the: ddATP, ddTTP, ddCTP, ddGTP. Each ddNTP is labeled with a different color of dye.
    - Because of this ingredient, the sanger sequencing method is also referred to as the chain termination method.

2.2.2.2. <u>Basic Methodology</u>

Sanger sequencing is a fairly simple technique. Generally, the process involves the following steps:

1. **Obtaining the Template Strand.**
   The first step involves obtaining the template strand from which the DNA can be sequenced, often from DNA amplification. This is done by applying heat to the gene which causes the double helix structure to denature, separating into two single strands. The single strand of interest will be used as the template strand.

[3] *(The Human Genome Project, n.d.) Source: https://www.genome.gov/human-genome-project*

2. **Addition of a Primer and dispersing into four (4) reaction tubes.**
   Once the temperature is lowered, the primer is added to bind onto the template strand. The primer will act as the mark where the sequencing will begin and extend from. The product formed is referred to as a prime DNA. The prime DNA is then dispersed into four reaction vessels or tubes.

3. **Addition of DNA Polymerase, Deoxy Nucleotides and Dideoxy Nucleotides into solution.**
   The next step involves the addition of the DNA polymerase and the four types of dNTPs, followed by a small proportion of one type of ddNTPs into each of the reaction tubes. This step initiates the sequencing phase. In each tube, the DNA polymerase slides along the template strand. As it slides, it reads each of the nucleotide bases, and from the primer, it inserts a complementary nucleotide, creating a complementary strand. However, the replication process is terminated each time a ddNTP is randomly incorporated in the growing complementary strand.

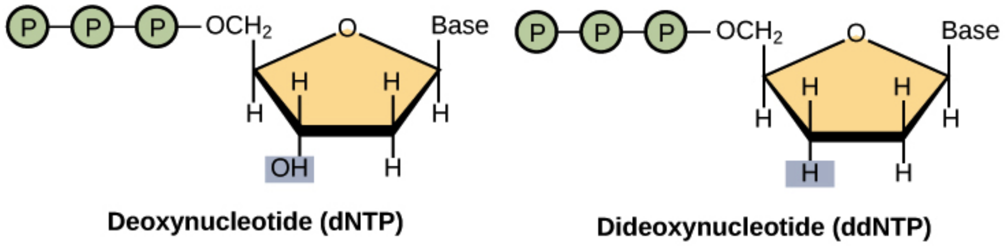

Figure SP-BT-TRB-2.2: A ddNTP is similar in structure to a dNTP, but lacks a 3' hydroxyl group (indicated by the box). When a ddNTP is incorporated into a DNA strand, DNA synthesis stops. (Source: https://openstax.org/books/biology/pages/17-3-whole-genome-sequencing).

   This is because ddNTPs are monomers with a missing hydroxyl group (–OH) at the site at which another nucleotide (dNTP) usually attaches to form a chain *(see Fig.SP-BT-TRB-2.2)*. Each ddNTPs is labelled with a different colored dye. This allows scientists to see them when they are exposed to UV light.

4. **DNA Replication and Termination.**
   In DNA replication, the previous step is repeated several times until no further nucleotides can be added. This would result in multiple strands of replicated DNA with varying lengths.

   In at least one reaction, it is virtually guaranteed that a ddNTP will have been incorporated at every single position of the target DNA. That is, the tube will contain fragments of different lengths, ending at each of the nucleotide positions in the original DNA *(see Fig.SP-BT-TRB-2.3)*. Since the ddNTPs are color dyed, they will indicate the ends of the DNA fragment.

5. **Denaturing the DNA fragments into single strand DNA**
   The DNA fragments obtained are then denatured into ssDNAs. This will prepare them for the next step called gel electrophoresis.

6. **Gel Electrophoresis**
   The final step involves separating pieces of DNA that are of different lengths by adding it to a gel base and exposing it to an electric current. This is referred to as gel electrophoresis.

Note

- *This step was covered in greater detail in Gr 11. Engineering (THEMA: BIOTECHNOLOGY – MODULE: Introduction to Biotechnology – Topic 3: Separation of biomolecules according to their charges. Refer students to their notes on this.*

The process causes the segments to travel through the gel according to their size. Essentially, the smallest pieces can move the farthest and the largest pieces do not move far at all. Once the pieces have finished moving, the gel is placed under an X-ray or a UV light, allowing scientists to see each segment.

To read the gel, you look at the dark bands in each column. There is one column for each type of nucleotide (G, C, A, T). By reading the sequence of the bands, you can determine the sequence of nucleotides.

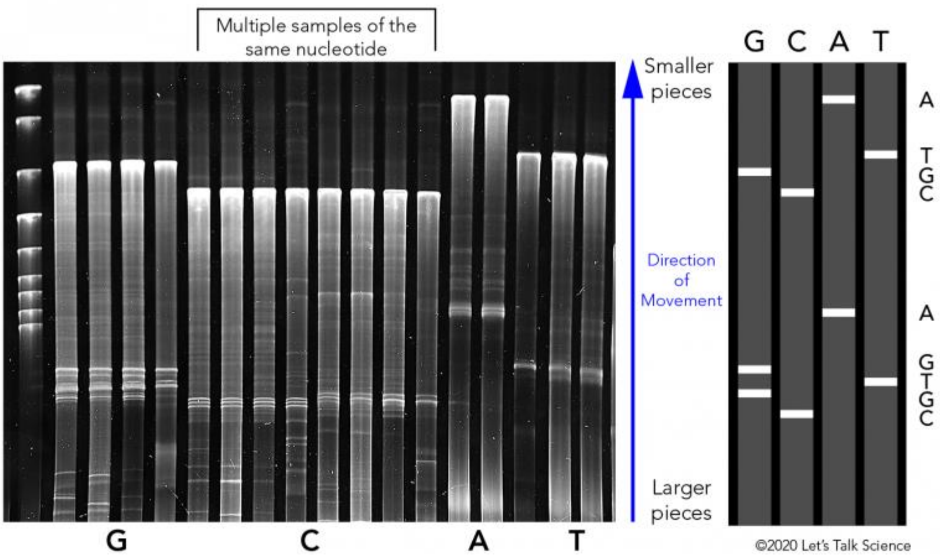

Fig. SP-BT-TRB-2.3: Left: X-ray that shows the columns and bands for the four nucleotides. Right: Bands and how they could be used to identify the order of the nucleotides (Source: Let's Talk Science using an image by vshivkova via iStockphoto).

Scientists have made changes to the original Sanger method. They use fluorophores. These are small chemical compounds that give off colored light. A different colored fluorophore is added to each nucleotide. In this way, sequencing can be performed in a single reaction mixture in a capillary tube.

To identify the sequence in the tube, scientists shine a laser through it. When the light passes through each band of color, a detector creates a peak on a graph. These peaks correspond to the nucleotide bases. By using fluorophores, scientists can automate the process of DNA sequencing. This means it can be done much more quickly. These steps are all demonstrated in the *Fig.SP-BT-TRB-2.4* below.

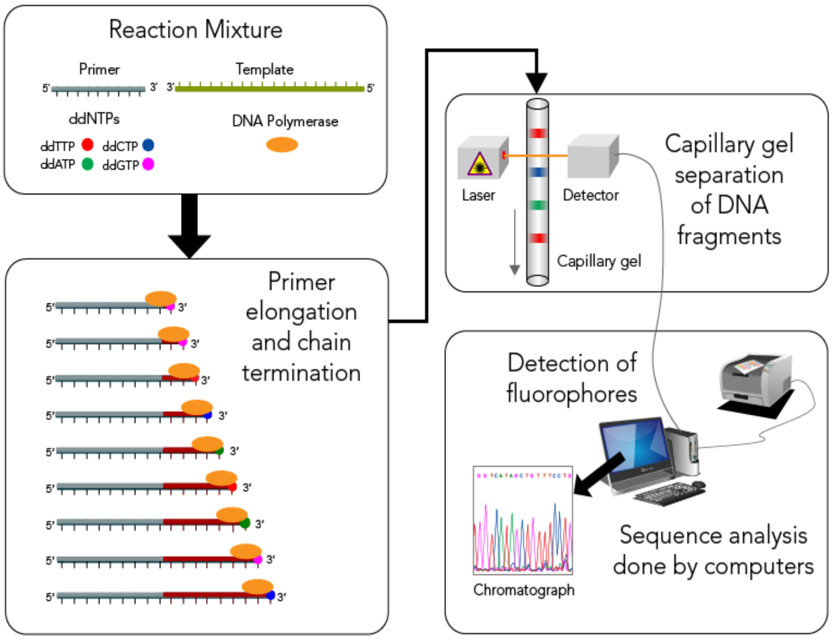

Fig. SP-BT-TRB-2.4: DNA sequencing using capillary gel electrophoresis (Source: Let's Talk Science using an image by Estevezj [CC BY-SA] via Wikimedia Commons).

2.2.3. Bioinformatics

As databases for gene and protein sequence grew at the end of the twentieth century, scientists turned to computers to help analyze this abundant and ever-growing amount of data. This resulted in the heavy use of bioinformatics, which is the science that collects biological data (such as gene and protein sequences) and stores them on computer databases.

Today, one of the most common tools used to examine DNA and protein sequences is the Basic Local Alignment Search Tool (BLAST). BLAST is a computer algorithm that is available for use online at the National Center for Biotechnology Information (NCBI) website, as well as many other sites. BLAST can rapidly align and compare a query DNA sequence with a database of sequences, which makes it a critical tool in ongoing genomic research.[4] Most of the sequences conducted can be compared and analyzed from the sequences stored on BLAST.

Based on the database, gene synthesis and cloning can be done. Gene synthesis is able to generate any new DNA sequence either naturally or artificially (without the need of a template DNA). Gene cloning is the method that replicates the original DNA, creating multiple clones.

[4] *(Ingrid Lobo, 2008)*

Activity 1: Laboratory Experiment

Title: DNA Extraction and Sequencing

This activity is designed to be an inter-phyla project with Biology. It is a laboratory experiment where students are tasked to perform DNA extraction and sequencing on three different specimens but of the same family:

- Viola odorata
- Viola ignobilis
- Viola occulta

Refer to the Gr 12. Biology SRB - THEMA: BIOTECHNOLOGY, MODULE: Bioinformatics for clear instructions.

Topic 3: Proteomics – Protein Sequencing

In this topic, the concepts of protein sequencing will be discussed. The objective is to relate the concepts of protein synthesis and cloning taught in **Chemistry** and the different techniques employed in **Physics** that biochemists use to identify the sequence of proteins. This will illuminate students to develop a broader understanding of the applications of physics in this field.

Key Points:

- Proteomics is the study that examines how proteins are expressed by the genome of an organism.
- Protein sequencing refers to the analysis and procedures involved in deciphering the sequence of the amino acid within a protein structure.
- Edman degradation is the technique used for sequencing the amino acids in proteins and peptides.

3.1. Introduction

Essentially, all proteins are made up of amino acids. Based on the genetic code stored in the gene, the amino acids are stringed together to form the primary structure of the protein. This is referred to as gene expression. When genes are expressed, they are synthesized into proteins. The whole set of proteins in cells is known as the proteomes. The properties, structures interactions, and roles of proteins are investigated using proteomics to study how proteins influence functions of the cell. Hence, Proteomics is the field of molecular biology concerned with the manner in which proteins are expressed by the genome of an organism.

Genomic studies provide crucial information for proteomic research since genes encode proteins. This is how both fields are complementary. Studies on proteomics are crucial in a variety of fields. This is especially relevant to the field of cancer biology. It can be utilized to identify abnormal proteins that cause cancer.

3.2. Gene Expression and Protein Sequencing

According to a specific ordering or expression of the amino acids, the protein will fold and it is based on this folding that will determine the protein's function. Higher stages of folding transform this primary structure into secondary, tertiary, and higher levels of structure. This means that the function of any protein is determined by the manner in which the amino acid is sequenced. The analysis and procedures involved in deciphering the sequence of the amino acid that constitutes the protein structure is referred to as protein sequencing.

The two major direct methods of protein sequencing are mass spectrometry and Edman degradation using a protein sequenator (sequencer). Mass spectrometry methods are now the most widely used for protein sequencing and identification but Edman degradation remains a valuable tool for characterizing a protein's N -terminus.

In this topic, we will focus on discussing the Edman Sequencing method.

Note
• *This topic will only highlight the concepts of Proteomics and Protein sequencing. The basics should be covered in greater detail in Gr 12. Chemistry (THEMA: BIOTECHNOLOGY - MODULE: 12.1 – Protein Sequencing). Refer students to their notes on this.*

3.2.1. Edman Degradation Process

Edman degradation is the technique used for sequencing the amino acids in proteins and peptides. This process sequentially removes one residue at a time from the amino end of a peptide without damaging the its entire structure.

Generally, the method involves labelling the n-terminal of the peptide that is located at the end of the amino sequence and then cleaving it from the parent bond. The labelled n-terminal amino acid is then identified using several techniques such as chromatography or electrophoresis.

The labelling of the n-terminal amino acid is done using a reagent called phenyl isothiocyanate and is referred to as Edman's reagent. The reaction between the Edman's reagent and the n-terminal amino acid done under mild alkaline conditions (around pH 8). This reaction gives rise to phenyl thiocarbamoyl derivative. Under heat and acidic conditions, this derivative is cleaved from the parent peptide and extracted by organic solvents. Once it is extracted, it is further acidified to form a stable cyclic compound of Phenyl thiohydantoin (PTH) amino acid. This makes it easier to identify using Chromatography techniques like High Performance Liquid Chromatography. This method can be repeated for the rest of the residues, separating one residue at a time. This process can be simplified in *Fig.SP-BT-TRB-2.1* below.

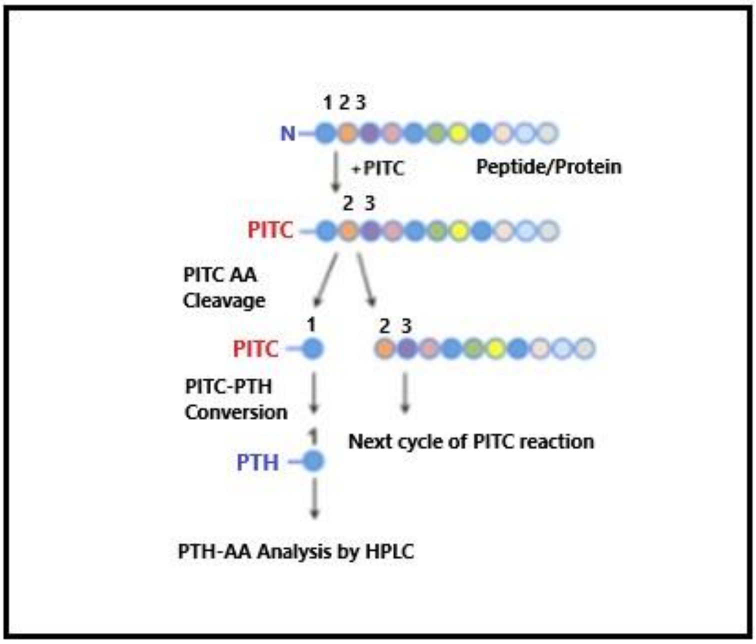

Figure SP-BT-TRB-2.1: Edman Degradation – Sample method

Edman degradation is very useful because it does not damage the protein. This allows sequencing of the protein to be done in less time. Edman sequencing is done best if the composition of the amino acid is known. To determine the composition of the amino acid, the peptide must be hydrolysed. This can be done by denaturing the protein, heating it and adding HCl for a long time. This causes the individual amino acids to be separated, and they can be separated by ion exchange chromatography. They are then dyed with ninhydrin and the amount of amino acid can be determined by the amount of optical absorbance. This way, the composition but not the sequence can be determined

While Sanger's method involved using enzymes to break the protein into various parts and then deducing the original sequence, Edman's method was more straightforward. There were no multiple methods involved; instead, there was just one step that was repeated over and over. Because the technique included starting at one end and breaking the protein step by step, it also gave the sequence directly.

Because of this efficiency, scientists like Stanford Moore and William Stein developed automatic protein sequencers (also known as sequenators) that could perform this procedure automatically.

One disadvantage of this approach is that it is unreliable for polypeptides (protein fragments) longer than 50 amino acids. As a result, large proteins are frequently broken down into tiny bits first (as in Sanger's approach) and subsequently degraded using Edman's method.

Similar to Genomics, BLAST is a significant tool used to rapidly align and compare a query of protein sequence with a database of sequences.

Activity 1: Exercise

Title: Using BLAST for sequence information

Link: ***https://teaching.healthtech.dtu.dk/teaching/index.php/Exercise:_BLAST***

This exercise was taken from the link provided above.

In this exercise, task the students to use BLAST for searching sequence databases such as GenBank (DNA data) and UniProt (protein). When using BLAST for sequence searches it is of utmost importance to be able to critically evaluate the statistical significance of the results returned.

The BLAST software package is free to use (Open Source) and can be installed on any local system — it is originally written for UNIX type Operating Systems. The package contains both programs for performing the actual sequence searches against preexisting databases (e.g., "blastn" for DNA databases and "blastp" for protein databases), as well as a tool for creating new databases from scratch (the "fortmatdb" program).

In this exercise we will be using the Web-interface to **BLAST hosted by the NCBI**. For our purpose there are several advantages to this approach:

- We don't have to mess around with a UNIX command prompt.
- NCBI offers direct access to preformatted BLAST databases of all the data that they host:

- GenBank (+ derivates)
- Full Genome database
- Protein database (Both from translated GenBank and UniProt)

It should be noted that running BLAST locally (for example at the super-computer cluster at CBS/DTU) offers much more fine-grained control of DATA and workflow (everything can be scripted/automated) than running BLAST through a web-interface.

Links

- NCBI BLAST main page: http://blast.ncbi.nlm.nih.gov/
- Notice: There are links to "Nucleotide BLAST" (including "blastn") and "Protein BLAST" (including "blastp") from this page.
- NCBI BLAST help pages

Part 1: Your first BLAST search

Below is the mRNA sequence for insulin from a South American rodent, the Degu (*Octodon degus*).

```
>gi|202471|gb|M57671.1|OCOINS Octodon degus insulin mRNA, complete cds
GCATTCTGAGGCATTCTCTAACAGGTTCTCGACCCTCCGCCATGGCCCCGTGGATGCATCTCCTCACCGT
GCTGGCCCTGCTGGCCCTCTGGGGACCCAACTCTGTTCAGGCCTATTCCAGCCAGCACCTGTGCGGCTCC
AACCTAGTGGAGGCACTGTACATGACATGTGGACGGAGTGGCTTCTATAGACCCCACGACCGCCGAGAGC
TGGAGGACCTCCAGGTGGAGCAGGCAGAACTGGGTCTGGAGGCAGGCGGCCTGCAGCCTTCGGCCCTGGA
GATGATTCTGCAGAAGCGCGGCATTGTGGATCAGTGCTGTAATAACATTTGCACATTTAACCAGCTGCAG
AACTACTGCAATGTCCCTTAGACACCTGCCTTGGGCCTGGCCTGCTGCTCTGCCCTGGCAACCAATAAAC
CCCTTGAATGAG
```

We will now use a BLASTN search at NCBI to determine whether this sequence looks like the human mRNA for insulin. There are two ways we can do this:

- search the entire database and look for human hits in the results,
- specifically search the human part of the database.

We will try both of these possibilities.

Search against NR

- Follow the "nucleotide blast" link from the main BLAST page.
- In the section "Program Selection" select the option "Somewhat similar sequences (blastn)"
- Choose "Nucleotide collection (nr/nt)" as the search database. NR is the "Non Redundant" database, which contains all non-redundant (non-identical) sequences from GenBank and the full genome databases.
- Click the BLAST button to launch the search.

After the search has completed, make yourself familiar with the BLAST output page. After a header with some information about the search, there are three main parts:

- **Graphic Summary**
  - each hit is represented by a line showing which part of the query sequence the alignment covers. The lines are colored according to alignment score.

-
- **Descriptions**
    - a table with a one-line description of each hit with some alignment statistics.
- **Alignments**
    - the actual alignments between the query and the database hits.

Note
• *You can toggle between hiding and showing each part by clicking on the part title (try it!).*

The columns in the Descriptions table are:

- Description — the description line from the database
- Max score — the alignment score of the best match (local alignment) between the query and the database hit
- Total score — the sum of alignment scores for all matches (alignments) between the query and the database hit (if there is only one match per hit, these two scores are identical)
- Query cover — the percentage of the query sequence that is covered by the alignment(s)
- E value — the Expect value calculated from the Max score (*i.e.,* the number of unrelated hits with that score or better you would expect to find for random reasons)
- Ident — the percent identity in the alignment(s)
- Accession — the accession number of the database hit.

First, take a look at the best hit. Since our search sequence (the query) was taken from GenBank which is part of NR, we should find an identical sequence in the search. Make sure this is the case!

QUESTION 1.1:

Answer the following questions about the best hit:

- what is the identifier (Accession)?
- what is the alignment score ("max score")?
- what is the percent identity and query coverage?
- what is the E-value?
- are there any gaps in the alignment?

Then, find the best hit from human (*Homo sapiens*) that is *not* a synthetic construct. (Tip: you can press Ctrl-F in most browsers to search in the page).

QUESTION 1.2:

Answer the same questions as before about the hit you found now.

Search against Human G+T

Note

- *In this context, G+T does not mean Gin and Tonic. Open a new window/tab with the BLAST home page. Make a new BLASTN search with the same query sequence, this time with Database set to Human genomic + transcript (Human G+T). Remember again to select Somewhat similar sequences (blastn) under Program Selection. Consider the best hit.*
- *Even though you may not have found exactly the same database entry in the two searches, the alignment should be the same. Make sure this is the case by comparing the actual alignments in the two windows where you made the searches.*

QUESTION 1.3:
Answer the same questions as before about the best hit you found in this search.

Concerning database size and E-values
When answering the previous two questions, you may have noticed that the E-value changed, while the alignment score did not. We will now investigate this further.

QUESTION 1.4:
What are the sizes (in base pairs) of the databases we used for the two BLAST searches? (**Tip:** Expand the "Search summary" section near the top by clicking it).

QUESTION 1.5:
i. What is the ratio between the database sizes in the two BLAST searches?
ii. What is the ratio between the E-values (for the best human hits) in the two BLAST searches?
iii. What is the relationship between database size and E-value for hits with identical alignment score?
iv. In conclusion: if the database size is doubled, what will happen to the E-value?

Note
• *For more activities similar to this, the above link can be used.*

MODULE 12.2: BIOELECTRONICS

4-5 Weeks

Bioelectronics utilizes electronic devices to monitor basic biological functions and is particularly used to assist biological systems in humans that are diagnosed to be defective or underperforming. These electronic devices help to modulate, initiate and trigger the response that will otherwise naturally be triggered by biological processes.

In this module, we will discuss some key applications of bioelectronics that are commonly used that will illuminate the mindset of students to investigate into how the concepts of physics can have biological applications.

Prerequisites:

- **NIL**

Bridging Phyla(s):

- **NIL**

Topic 1: Applications of Bioelectronics

Bioelectronics is basically the use of electronic devices to monitor basic biological functions. In this topic, we will discuss some of the electronics utilized to assist the biological functions that are deficient or underperforming in humans/people with certain medical conditions.

Key Points:

- Physics is applied through the creation and use of electronic devices that help monitor basic biological functions
- Cardiac Pacemakers artificially generate electro-impulses to stabilize heart rates and are used by patients experiencing cardiac problems.
- Hearing Aid is a bioelectronics device that enables persons experiencing hearing losses or have hearing impairments to hear properly.

1.1. Introduction

1.1.1. Definition

Bioelectronics is the branch of science concerned with electronic control of physiological function, especially as applied in medicine to compensate for defects of the nervous system.[5]

Essentially, bioelectronics is the application of the principles of electronics to biology and medicine. Many science-fiction novels and movies have expanded the horizons with its imagination such as Star Wars, iRobot and Terminator, which have shown us what modern technology is or can be capable of doing. There are several applications that display the potential use of bioelectronics and can range from artificial limbs, humanoids, and the various sensors (biosensor) that are attached to the body.

In this topic, we will discuss two common applications of bioelectronics.

1.2. Applications of Bioelectronics

1.2.1. Cardiac Pacemaker

The pacemaker is an electrical pulse generator that creates electro-impulses, that would normally be generated by the heart, to trigger the contractions of the heart muscles to artificially stabilize the beating rates. Thus, it is used when the natural cardiac system fails or is failing to pump blood at a normal rate.

The workings of the pacemaker can be understood by first understanding the natural processes that occur in the heart.

1.2.1.1. Basic Anatomy and Electrophysiology of the Heart

The heart is the engine of the body, pumping blood through the circulatory system and supplying the body with oxygen and nutrients. The heart consists of four chambers: The left and right atrium and ventricle respectively, as depicted in *Fig.SP-BT-TRB-4.1*.

[5] *(Merriam-Webster.com, 2022)*

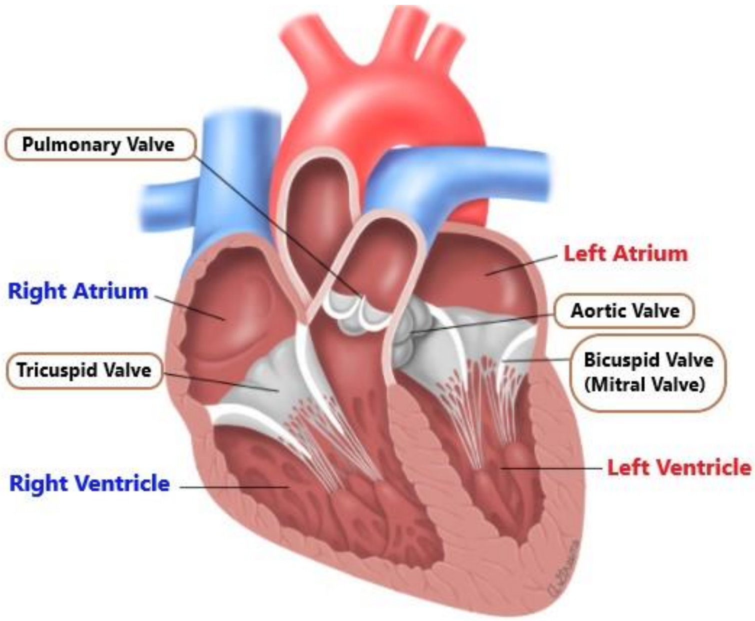

Figure SP-BT-TRB-4.1: Basic anatomy of the human heart. The chambers heighted blue and red represent the deoxygenated and oxygenated blood respectively. (Modified Source: Bing images - heart_chamber_anatomy_pi_utd.jpg (geniemd.net)]).

On each side, the atrium is connected to the ventricle by a one-way valve called the bicuspid (left side) and tricuspid (right side). Blood is pumped as these chambers contract and relax in turn. The pumping motions is controlled and regulated by electrical impulses distributed in the heart by a conduction system called the cardiac conduction system.

The process begins as the atria fills with deoxygenated blood from the body (blue on the left) and oxygenated blood (red on the right) *(see Fig.SP-BT-TRB-4.2)*. An electrical signal from the Sinoatrial (SA) node then causes the atria to contract, forcing the blood into the ventricles via the tricuspid valve. The electrical signal is then picked up by the Atrioventricular (AV) node and directed into the bundle of His and down to the purkinje fibres in the ventricle walls. This causes the ventricles to contract. Subsequently, the blood is pumped into the pulmonary valve (on the right) to the lungs and the aortic valve (on the left) to the rest of the body. These valves close and the cycle is then repeated.

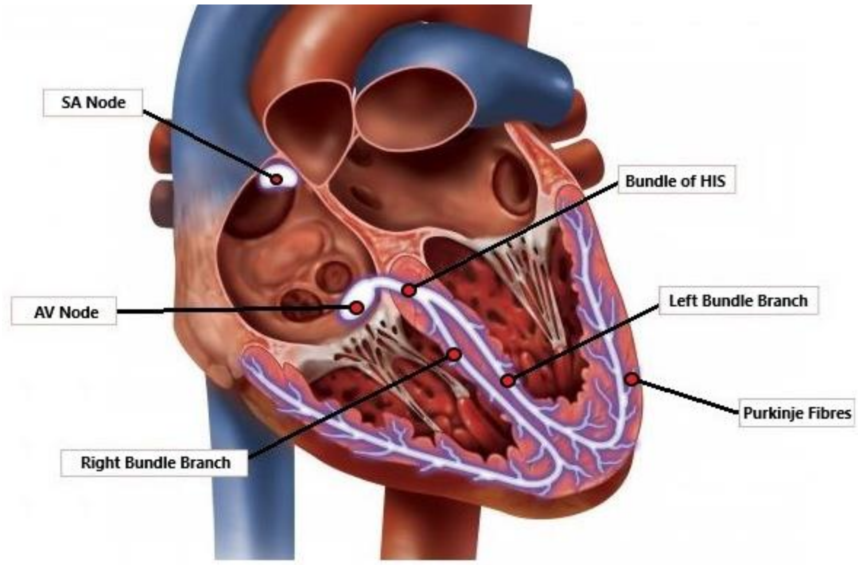

Figure SP-BT-TRB-4.2: Basic Electrophysiology of the human heart (Modified Source: Bing images - heart_chamber_anatomy_pi_utd.jpg (geniemd.net)).

The SA is the natural pulse generator or pacemaker of the heart and determines how often the heart beats per minute (bpm). At rest, a healthy heart beats around 60 – 80 bpm. If the heart beat drops to consistently less than 60 bpm, this is referred to as Bradycardia. This slowed heart rate does not permit a sufficient amount of oxygen to be circulated and can lead to dizziness and/or fainting.

If the SA node fails to function correctly, an artificial pacemaker, referred to as cardiac pacemaker, can be fitted to help regulate the heart beat with small evenly timed electric shocks. If the electrical signal of the heart is below the pacemaker's programmed rate, then the pacemaker resets the electrical impulses to the normal rate.

1.2.1.2. Mechanism of the Cardiac Pacemaker

The cardiac pacemaker contains a small computer and a battery which is programmed to help monitor the heart rates. It is implanted through a small incision below the collar bone just under the skin and connected to the heart by wires called "leads" through a selected blood vessel. Most pacemakers use two leads and are connected to the right atrium and right ventricle respectively *(see Fig. SP-BT-TRB-4.3)*. Usually, the location of the pacemaker's pulse generator is positioned on the side that is least used by the patient. Thus, if they are right-handed, it will go on the left side of the collarbone and vice versa.

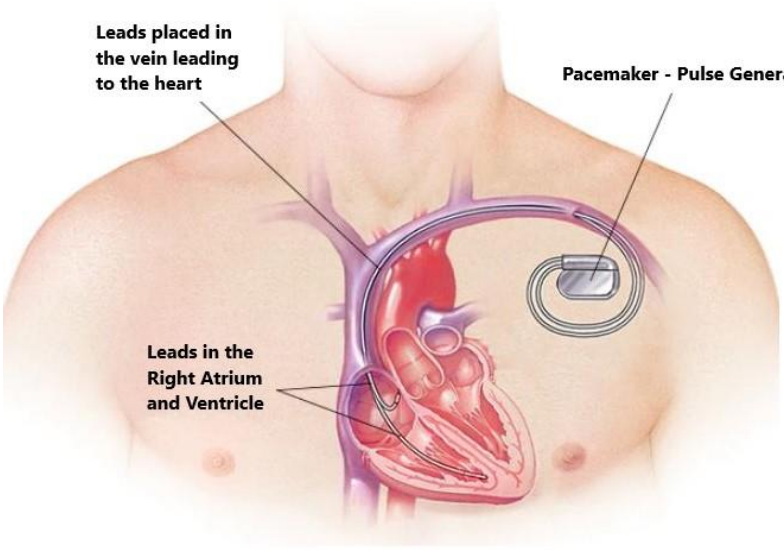

Figure SP-BT-TRB-4.3: Placement of the Pacemaker. This is also similar to how the ICD is implanted. (Modified Source: Bing images - https://www.mayoclinic.org/-/media/a68e050675aa48b6a215bcd104048ca7.jpg)

Through the leads, the pacemaker can monitor the heartbeat and treat rhythm disturbances specifically. If the heart rate drops or increases significantly, it emits electrical pulses to the heart. The pulses from the pacemaker, will normalizes and thus stabilizes the heart rhythm.

For conditions where the rhythm of the ventricles is dangerously abnormal, an Implantable Cardioverter Defibrillator (ICD) can be used. It is capable of sensing life threatening heart rates or even a stopped heart and sends an electrical signal powerful enough to restart it. This is the main difference between the pacemaker and ICD. An ICD can also function as a pacemaker where it sends out a signal when the heart rate is gets too slow.

For some conditions, an even more sophisticated device called a Cardiac Resynchronization Therapy (CRT) ICD is implanted. This device uses a third lead inserted into the left ventricle to resynchronize the ventricles when necessary.

However, the implanting of these leads from all the devices can cause problems of their own. Some of the patients with ICDs can have lead failures within 10 years. In most cases, this would then require open heart surgery. This has resulted on technologies that have created several pacemakers that do not rely on placing leads inside the heart. One design, the Subcutaneous IDC, places the lead just outside the heart, under the patient’s skin. Wireless designs are also now being designed that may eventually phase out the leads altogether.

1.2.2. Hearing Aid

Another bioelectronic device is the Hearing-Aid that makes sound more audible for persons experiencing hearing losses or have hearing impairments. It can also assist people in hearing better in both quiet and noisy situations.

We can understand the mechanism of the hearing aid by first understanding how different parts of a normal and functional ear work.

1.2.2.1. How the Ear works

The ear is constantly active body part that can pick up vibrations of particles or sound waves from the surrounding environment and convert them into information that the brain can interpret, such as speech or music. The sound waves can either vibrate slowly or rapidly to produce different effects: slow vibrations produce deeper sounds and faster vibrations produce high-pitched sounds.

There are three main sections of the ear. The Outer, Middle and Inner Ear *(see Fig. SP-BT-TRB-4.4)*.

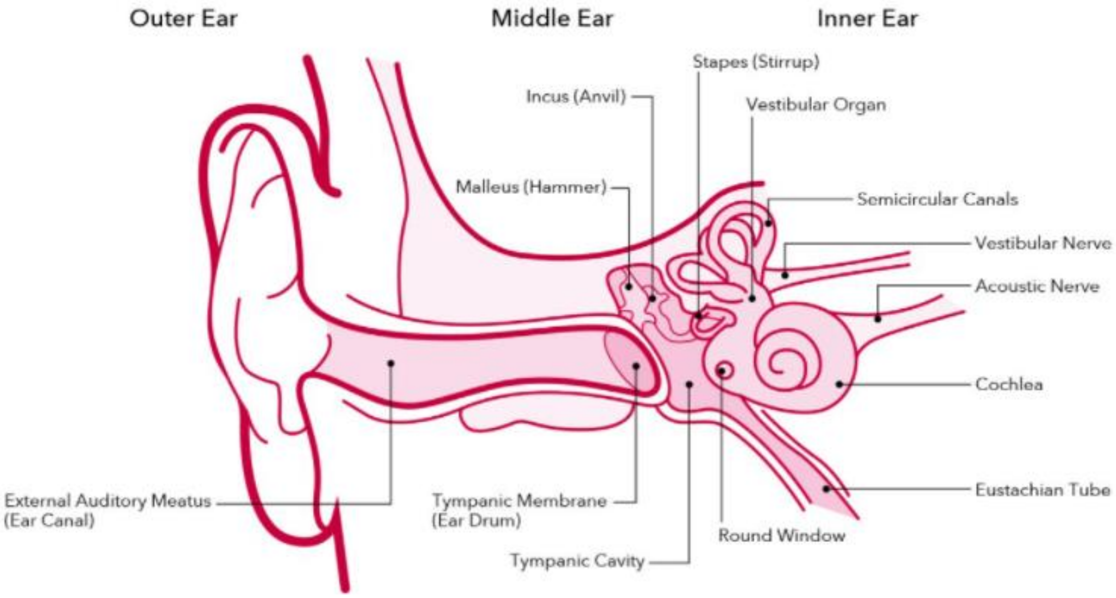

Figure SP-BT-TRB-4.4: Anatomy of the Ear which consists of the Outer, Middle and Inner Ear. (Modified Source: https://www.amplifon.com/au/hearing-loss/how-does-the-ear-work/parts-of-the-ear)

1. Outer Ear

Sound is first collected by the visible part of the outer ear called the auricle (sometimes referred to as pinna). From the auricle, the sound waves are channeled through the ear canal (or external auditory meatus), where the sound is amplified. At the end of the canal, the sound waves come into contact with a flexible, oval membrane called the eardrum (or tympanic membrane). This causes the eardrum to vibrate.

2. Middle Ear

The vibrations then activate the middle section of the ear, referred to as the ossicles, forcing them into motion. Ossicles are actually three different bones, and are the smallest in the human body. These three bones are named after their shapes: the malleus (hammer), incus (anvil) and stapes (stirrup). The ossicles further amplify the sound.

The pressure between the air outside and within the air is equalized by the Eustachian tube, located at the entrance of the middle ear. The amplified sound exits the stapes and connects to the oval window that connects the middle ear to the inner ear.

3. Inner Ear

From the oval window, the sound waves enter the inner ear and then into a snail-shaped organ called the cochlea. The cochlea is filled with a fluid that moves in response to the vibrations from the oval window. As the fluid moves, thousands of nerve endings (about 25,000) are set into motion. These nerve endings transform the vibrations into electrical impulses that then travel along the eighth cranial nerve (auditory nerve) to the brain. The brain then interprets these signals, and this is how we hear. The inner ear also contains the vestibular organ that is responsible for balance.

1.2.2.2. Mechanism of the Hearing Aid

A hearing aid is like a mini-PA system that has three basic parts: a microphone, amplifier, and speaker. The hearing aid receives sound through the microphone. Most hearing aids use two microphones to help cancel out background noise. The microphone performs two functions; it captures the sound and converts them into electric signals.

The sound-to electric conversion occurs within a section of the microphone containing the diaphragm. This is a small and thin plastic that acts like the eardrum, moving back and forth as the sound waves come into contact. A coil attached to the diaphragm and a permanent magnet is also forced to oscillate, generating electric current or signals through electromagnetic induction. The electric signals then flow to an amplifier that then increases the power of the current and then sends them through the ear to the speaker.

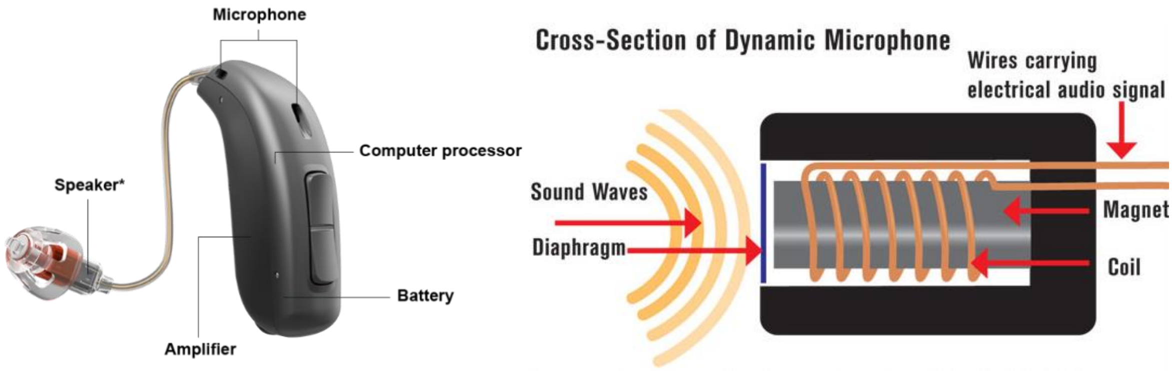

Figure SP-BT-TRB-4.5: (Left) Basic parts of a hearing aid; (Right) Basic mechanism of a microphone. (Modified Source: https://www.videomaker.com/article/c4/15355-microphone-anatomy)

The electric signals are then sent to an Analog to Digital Converter (ADC). As its name suggests, it will convert the electric current into digital signals which will then be processed by the hearing aids' computer chip. This chip is programmed according to the person's hearing threshold. Based on threshold, the chip will amplify the incoming sound to accommodate the person's hearing loss. Once it is amplified, the digital signal is converted back into analog via a Digital to Analog Converter (DAC). The amplified signals are then sent to the speaker where it is played into the hearing aid user's ear.

Activity 1: Research Assignment

In this activity, you will task the students to research and identify at least two other bioelectronic devices. From their research, they must explain what biological system is dysfunctional or experiencing problems that prompts the usage bioelectronics.

The students should also provide illustrations or diagrams that they have constructed to assist in their explanation of the mechanism of their chosen bioelectronic device.

THEMA: HEALTH AND MEDICINE

Introduction

The study of the human mind and body, how they function, and how they interact - not only with one another but also with their surroundings - has always been critical to human well-being. With the advancement of modern medicine, research on new therapies and preventative medicine has exploded, and a network of disciplines, including genetics, psychology, and nutrition, is working to improve human health.

MODULE 12.1: NUCLEAR MEDICINE

4-5 Weeks

Nuclear medicine is the science and clinical practice that involves the usage of radioactive compounds called radiopharmaceuticals (or more commonly called radiotracers) that are administered into the body. The radiation emitted is captured by certain medical instruments and assist nuclear medical practitioners to identify different types of diseases that can range from cancer, heart diseases and neurological disorders.

In this module, we will discuss the Physics associated with Nuclear Medicine. We will begin by understanding the nucleus of atoms at the sub-atomic level and how it relates to certain elements and chemical compounds becoming radioactive. This will serve as the basis to understanding radiotracers used in nuclear medicine and will assist students to understand how these instruments administer and use radiotracers to help identify and prevent the spread of the medical conditions mentioned above.

The main focus of Topic 1 is on Radionuclides used in Nuclear Medicine. Topic 2 focusses on Imaging modalities (SPECT and PET Scans). Topic 3 focusses on Therapeutic applications emphasizing on serious treatments such as cancer treatment and bone metastasis.

Prerequisites:

- Grade 11 Term 3, STEM Physics (Thema: Health and Medicine)

Bridging Phyla(s):

- NIL

Topic 1: Introduction to Nuclear Medicine

In this topic, we will discuss the basic practices of nuclear medicine in the broader field of medicine and radiology.

Key Points:

- Nuclear medicine involves administering small amounts of radiopharmaceuticals in the body for diagnosing diseases.
- Radiopharmaceuticals emit radiation that are captured by nuclear imaging instruments that display an image containing medical information for diagnostics.
- From diagnostics, if medical practitioners detect serious medical conditions that can be prevented will advise for the patient to undergo nuclear therapy.
- Two common modalities used in diagnostics are PET and SPECT imaging.
- Radiopharmaceuticals are made artificially either in research nuclear reactors or particle accelerators called cyclotrons.

1.1. Understanding Nuclear Medicine

Nuclear medicine involves the use of radioisotopes, called radiopharmaceuticals, radionuclides or radiotracers, administered into the body in very small amounts to examine organ structure and functionality. As such, it is a specialized field of radiology.

Nuclear medicine is a multidisciplinary branch of medicine (mainly Chemistry, Physics, Biology, and Computer Technology) but is more closely related to radiology, medical imaging and diagnosis. In particular, it is often used to help diagnose, evaluate and treat a variety of diseases or abnormalities that are very early in the progression of a disease. Some examples include neurological disorders and cancers such as liver, thyroid and prostate cancer. *Fig.SP-HM-TRB-1.1* illustrates the radiotracers used for diagnosis and therapy for some of the common diseases.

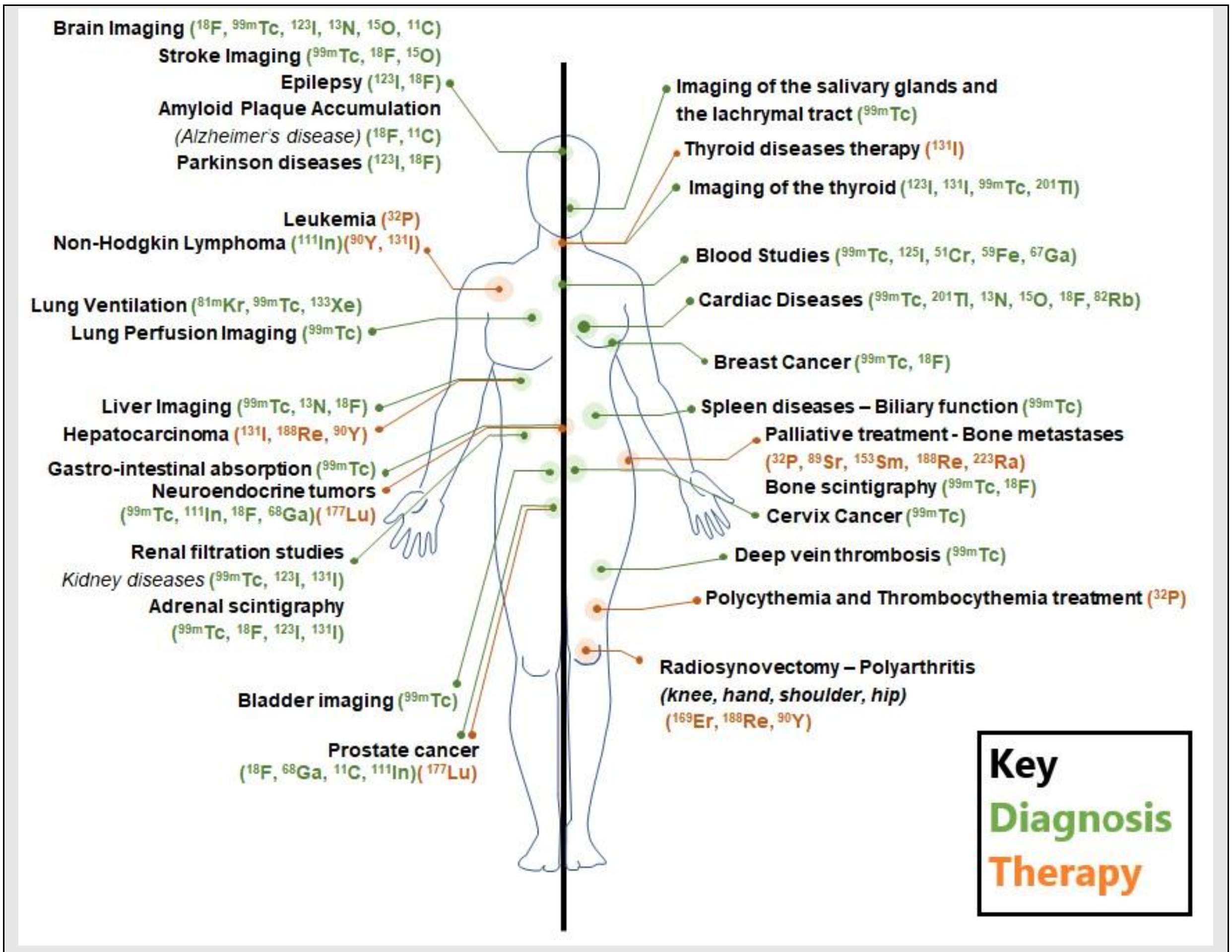

Figure SP-HM-TRB-1.1: Use of the different radiotracers that are administered into the body for diagnosis and therapy. (Modified Source: Nuclear Medicine Europe - The Industry Association)

1.2. How Nuclear Medicine Works

Firstly, a small amount of a certain radiopharmaceutical is ingested (either injected, swallowed, or inhaled as a gas) through the patient's body. Radiopharmaceuticals are radioactive isotopes of different elements *(as displayed in Fig.SP-HM-TRB-1.1)* and therefore emit radiation. When introduced into the system, the radiotracer will have eventually accumulated in the organ or body part that is to be diagnosed or treated. The radiations emitted from the radiotracer are then detected by special cameras in the imaging machines that will provide in-depth pictures and useful molecular information. This biological information is usually done as a diagnosis to detect a wide range of diseases (some of which were mentioned previously). The process of obtaining the biological information from these instruments is referred to as nuclear imaging, and is used primarily for diagnostic nuclear medicine.

As opposed to conventional x-rays, where the radiation source is external to the patient, nuclear medicine and imaging uses internal radiation from the decay of the radiotracer within the patient's body to capture an image.

Nuclear medicine plays a pivotal role in medical diagnostics and therapeutics. Diagnostic nuclear medicine mainly deals with PET (Positron Emission Tomography) and SPECT (Single-Photon Emission Computed Tomography) imaging. In many cases,

these scans are combined with CT (Computed Tomography) tests in order to create a scan that is more accurate and precise.

Note

- *CT scan was covered in Physics Gr 11: THEMA: Biotechnology; MODULE: Medical Physics; Topic 3: Computed Tomography (CT) Scan.*
- *PET and SPECT will be discussed more in Topic 2: Nuclear Medicine Diagnostics.*

Usually following the medical diagnosis is therapy. Therapeutic nuclear procedures include radioactive Iodine (I-131 or ^{131}I) therapy and radio-immunotherapy. Radio-immunotherapy combines radiation therapy and the targeting ability of immunotherapy. Generally, this involves stimulating or suppressing the body's immune system, so the radiation is then able to target and fight the affected areas very precisely. Thus, therapeutic nuclear medicine is typically used for severe types of cancers and diseases.

1.3. Radiopharmaceuticals

Radiopharmaceuticals are developed on the basis of observing the biological activities that occur within the human body, specifically in how certain organs are capable of absorbing chemicals. For example, the thyroid takes up iodine and the brain consumes glucose. Based on these observations, most radiopharmaceuticals are developed by attaching radioisotopes to biomolecules. This will allow the radiotracer to enter the body and be incorporated into its biological pathways, metabolized, and then excreted.

Several different types of radiopharmaceuticals are available and most are isotopes of different elements. They are often made in a nuclear reactor or through the particle accelerators such as a cyclotron. In both cases, they expose a target material to sub-atomic particles (such as protons or neutrons) to stimulate a nuclear reaction and then place it through several different chemical process to produce a desired radioisotope.

Provided below are two tables that display the different medical radioisotopes that are artificially manufactured using a reactor and a cyclotron.

Table SP-HM-TRB-1.1: Reactor-produced Medical Radioisotopes

Radioisotope	Half-life	Use
Phosphorus-32	14.26 days	Used in the treatment of excess red blood cells.
Chromium-51	27.70 days	Used to label red blood cells and quantify gastro-intestinal protein loss.
Yttrium-90	64 hrs	Used for liver cancer therapy.
Molybdenum-99	65.94 hrs	Used as the 'parent' in a generator to produce technetium-99m, the most widely used radioisotope in nuclear medicine.
Technetium-99m	6.01 hrs	Used to image the brain, thyroid, lungs, liver, spleen, kidney, gall bladder, skeleton, blood pool, bone marrow, heart blood pool, salivary and lacrimal glands, and to detect infection.
Iodine-131	8.03 days	Used to diagnose and treat various diseases associated with the human thyroid.

Samarium-153	46.28 hrs	Used to reduce the pain associated with bony metastases of primary tumors.
Lutetium-177	6.65 days	Currently in clinical trials. Used to treat a variety of cancers, including neuroendocrine tumors and prostate cancer.
Iridium-192	73.83 days	Supplied in wire form for use as an internal radiotherapy source for certain cancers, including those of the head and breast.

(Source: Radioisotopes | What are Radioisotopes? | ANSTO)

Table SP-HM-TRB-2.2: Cyclotron-produced Medical Radioisotopes

Radioisotope	Half-life	Use
Carbon-11	20.33 mins	Used in PET scans to study brain physiology and pathology, to detect the location of epileptic foci, and in dementia, psychiatry, and neuropharmacology studies. Also used to detect heart problems and diagnose certain types of cancer.
Nitrogen-13	9.97 mins	Used in PET scans as a blood flow tracer and in cardiac studies.
Oxygen-15	2.04 mins	Used in PET scans to label oxygen, carbon dioxide and water in order to measure blood flow, blood volume, and oxygen consumption.
Fluorine-18	1.83 hrs	The most widely-used PET radioisotope. Used in a variety of research and diagnostic applications, including the labelling of glucose (as fluorodeoxyglucose) to detect brain tumors via increased glucose metabolism.
Copper-64	12.7 hrs	Used to study genetic disease affecting copper metabolism, in PET scans, and also has potential therapeutic uses.
Gallium-67	78.28 hrs	Used in imaging to detect tumors and infections.
Iodine-123	13.22 hrs	Used in imaging to monitor thyroid function and detect adrenal dysfunction.
Thallium-201	73.01 hrs	Used in imaging to detect the location of the damaged heart muscle.

(Source: Radioisotopes | What are Radioisotopes? | ANSTO)

1.3.1. Technium-99m

Technetium-99m (Tc-99m) is a nuclear isomer of Tc-99 and is one of the most commonly used radiopharmaceuticals for imaging and diagnostic purposes for organs and tissues. It is often favored amongst other radiopharmaceuticals because its use is not limited to a primary organ and that it has a short six-hour half-life.

1.3.1.1. Production

A large majority of the supply of Tc-99m comes from Technetium generators, which are located in nuclear research reactor plants. These generators contain Molybdenum (^{98}Mo) and facilitate a process called nuclear bombardment. Basically, this involves firing the nucleus of $^{98}_{42}Mo$ with neutrons resulting in the creation of Molybdenum-99 ($^{99}_{42}Mo$). The added neutron will create an unstable nucleus and will cause ^{99}Mo to reach a lower energy level to become stable. This process is called radioactive decay and it will disintegrate to half of its life (half-life, $T_{1/2}$ - 66 hours) to produce Tc-99m.

The production of Tc-99m is then able to be used for medical purposes. Tc-99m is also unstable and thus undergoes another radioactive decay to produce Ruthenium-99 (see *Fig.SP-HM-TRB-1.2)*.

$$^{98}_{42}\mathrm{Mo} + ^{1}_{0}\mathrm{n} \rightarrow ^{99}_{42}\mathrm{Mo} \qquad ^{99}_{42}\mathrm{Mo} \rightarrow ^{99}_{43}\mathrm{Tc} + ^{0}_{-1}\beta$$

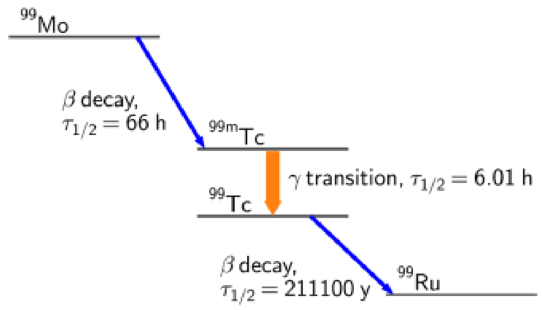

Figure SP-HM-TRB-1.2: Nuclear reaction equation for the production of Technetium-99m and an illustration of the overall radioactive decay (beta decay) with their respective half-life's. (Source: Production of Technetium-99m - Radioactive Isotopes (weebly.com))

Note

- *The concepts of radiation, radioactive decay and half-life will be covered in detail in Gr 12: THEMA: ENERGY; MODULE: Nuclear Energy.*

1.3.1.2. Administration and Mechanism of Action

Technetium radiotracers may be administered via injection intravenously or orally. They exert their effects by emitting gamma rays, which are then picked up by a gamma camera for imaging. The radiotracers are not localized to a primary organ and distribute equally to multiple tissues. Once distributed, they emit photons that can be captured for imaging with SPECT/CT or PET/CT. This can display imaging of different organs and tissues such as the brain, bones, lungs, kidneys, thyroid, heart, gall bladder, liver, spleen, bone marrow, salivary and lachrymal glands, blood pool, and sentinel nodes.

Tc-99m's short half-life accounts for less total radiation exposure to the patient. Furthermore, the region of perfusion in an organ or tissue is evaluated by the uptake of the radiotracer, determining reversible or irreversible ischemia. *Table SP-HM-TRB-1.3* displays some of the different isotopes of Tc-99m approved by the Food and Drug Administration (FDA) for clinical use.[6]

Table SP-HM-TRB-1.3: FDA Approved Technetium-99m Use

Technetium-99m Isotopes	**Use**	**Scintigraphy (Scan)**	**Radiation level (MBq)**
Tc-99m Sodium Pertechnetate	Used for diagnostic imaging of the thyroid, salivary gland, urinary bladder, vesicoureteral reflux, and nasolacrimal drainage imaging.	Brain scintigraphy (adult)	370 - 740
		Thyroid scintigraphy (adult)	37 - 370
		Salivary gland scintigraphy (adult)	37 - 185
		Blood pooling scintigraphy (adult)	370 - 1110

[6] *(Steven M. Kane and Donald D. Davis, 2021)*

	Its use in the gastrointestinal tract is primarily for the diagnosis of Meckel's diverticulum (Meckel scintigraphy scan).		
		Vesic-ureteral scintigraphy (adult)	18.5 - 37
		Nasolacrimal drainage scintigraphy (adult)	3.70
		Brain scintigraphy (children)	5.2 - 10.4
		Thyroid scintigraphy (children)	2.2-3.0
		Blood pooling scintigraphy (children)	5.2-10.4
		Vesic-ureteral scintigraphy (children)	18.5 - 37
Tc-99m Sestamibi	Used for Cardiac perfusion imaging, assessing function, and localizing ischemia and infarction - also used for breast imaging.	Myocardial perfusion scintigraphy	555 - 1110
		Parathyroid surgery preparation	740 - 925
		Breast scintigraphy	740-1110
Tc-99m Tilmanocept	Used to localize lymph nodes drainage of primary tumors.	Lymph node biopsy/scintigraphy	18.5
Tc-99m Exametazime	Used for imaging abdominal infections, inflammatory bowel disease, and brain perfusion.	Cerebral flow scintigraphy	370-740
		Leukocyte labeled scintigraphy	259 - 925
Tc-99m Macroaggregated Albumin	Used to assess pulmonary perfusion.	Lung scintigraphy	37-148
		Porto venous shunt	37-111

NB: *Technetium (Tc99m) radiotracer may be administered via injection intravenously or orally. Dosages may be modified depending on the patient and indications of imaging. (Source: Technetium-99m - StatPearls - NCBI Bookshelf (nih.gov))*

1.4. Advantages and Disadvantages of using Nuclear Medicine

I. Advantages of Nuclear Medicine

1. Allows for an accurate diagnosis

Complex medical procedures have become much safer now thanks to the technologies of nuclear medicine. Instead of exploratory surgeries, a detailed internal image can be taken instead. This enables the process for diagnosis to be simpler and relatively safe as it eliminates some of the dangerous procedures performed in the past.

2. Permits an early disease detection

For diseases like cancer, early detection is critical for a patient to receive a high likelihood of long-term survival. Nuclear medicine makes it possible to find early-stage cancers and other illnesses or health problems so that an effective treatment plan can be developed.

3. Provides a treatment alternative

Apart from diagnosing health problems, nuclear medicine is also used to cure some of the diseases and/or send them into remission. Radiation treatments, chemotherapies, and other cutting-edge options give patients a new hope for life when none in the past would have existed.

II. Disadvantages of Nuclear Medicine

1. It is Very Expensive

Nuclear medicine is quite expensive on all fronts. This is because of the technology behind the therapies or diagnostic equipment that have high installation and maintenance costs. These costs are passed along to each patient so the medical provider can make money.

2. There are Possible Health Risks

Because of the radiation that can be contracted due to consistent exposure, certain health issues can arise as a side effect of using nuclear medicine. This is especially true in infants, toddlers, the elderly, and women who may be pregnant. As such, most doctors recommend an alternative treatment if possible.

3. It is Not Guaranteed to be 100% Effective

Although nuclear medicine offers some of the best medical treatment options available today, it still isn't a fool-proof system. No medical procedure is guaranteed to work 100% of the time.

For many people, the main disadvantage of nuclear medicine is the cost of it. If the costs can be managed, then the benefits almost always outweigh the other disadvantages which may be present.

Activity 1: Review Exercise

Answer the following questions:

Questions

1. Define the following terms:

a. Nuclear Medicine

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______

b. Radiopharmaceutical

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______

c. Medical Imaging

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2. What are SPECT and PET scans?

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3. What are the different ways in which nuclear medicine is administered?

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4. Explain how radiopharmaceuticals does not immediately expose patients to radiation. Also articulate and explain why how it is developed to be used in the human body?

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5. List at least three different radiopharmaceuticals, their half-lives and medical use.
Sample Answer

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6. List one major advantage and disadvantage of using nuclear medicine.

Topic 2: Nuclear Medicine - Diagnosis

This topic will discuss the processes involved in nuclear medicine diagnostics and the basic working mechanism of the SPECT and PET that are used with CT scans.

Key Points:

- Nuclear Medicine Diagnostics is the most common technique used that involves administering radiopharmaceuticals to diagnose different medical conditions.
- Nuclear medicine uses Emission Tomography which are of two types:
  - Single Photon Emission Computed Tomography – emission of gamma rays from the radiotracers within a patient's body and diagnoses tissue function.
  - Positron Emission Tomography – captures gamma radiation from a positron.
- In SPECT scans:
  - It displays how blood flows into and within tissues and organs. This will help to diagnose and monitor mostly brain, heart or bone-related disorders.
  - Any radiotracer that emits gamma ray photons are used in SPECT scans.
- In PET scans:
  - It displays how blood flows into and within tissues and organs. This will help to diagnose and monitor mostly brain, heart or bone-related disorders.
  - Positrons ($\beta+$) are emitted immediately once the radiotracer is introduced to the body and has a very short half-life.
  - Positrons travel a small distance and annihilate when they interact with electrons in the tissue.
  - This annihilation produces a pair of gamma-ray photons which travel in opposite directions.

2.1. Introduction

Generally, radioisotopes that are used in nuclear medicine have short half-lives, which means they decay quickly and are suitable for diagnostic purposes. Those that have relatively longer half-lives take more time to decay and this makes them suitable for therapeutic purposes. In diagnosis, most of the scans involve Emission Tomography.

Note

- *We will discuss more on atomic structure, fundamental particles, radiation, radioactive decay and half-life in Gr 12 THEMA: ENERGY*

2.1. Emission Tomography

Emission tomography is a medical image modality that provides functional and/or metabolic information about an organ, a system or tissue of a biological organism. This is achieved by administrating radionuclides to the patients and imaging the emitted radiation. The acquired information is useful not only for diagnostic purposes, such as detection of functional abnormalities or early identification of tumors, but can also provide significant assistance in follow-ups and therapy planning.

Discussed previously, there are two types of emission tomography: SPECT - Single Photon Emission Computerized Tomography, where the nucleus eliminates its energy

by emitting one or more photons, and PET, Positron Emission Tomography, where the excess nuclear energy is emitted with a positron (anti-particle of an electron). The tomograms are reconstructed from the external detection of the emitted photons or the pair of annihilation photons in coincidence. Both differ depending on the decay mode of the chosen radionuclide.

2.2. Single Photon Emission Computerized Tomography

SPECT imaging uses a radiotracer that is administered and distributed within the patient's body and Computed Tomography to construct three-dimensional (3D) images. This is achieved by using a special gamma camera detector that can detect and capture the gamma-rays emitted from the radiotracers at different angles. Gamma rays are a type of electromagnetic wave that have shorter wavelengths than visible light but travel faster (except in a vacuum where all electromagnetic waves travel at the same speed). The cameras are mounted on a rotating gantry that allows the detectors to be moved in a tight circle around a patient who is lying motionless on a pallet *(see fig. SP-HM-TRB-2.1)*. The data collected from many angles allows cross-sectional images of the distribution of the radionuclide to be reconstructed, thus providing the depth information missing from planar imaging which provides little depth information.

Figure SP-HM-TRB-2.1: Image of a SPECT scan where the patient lies on the pallet and the two gamma cameras that are angled and situated on the main scanning frame and rotate to obtain cross-sectional images that are combined to construct 3-D images. (Source: Bing Images)

Difference to other Imaging Techniques

The most common imaging techniques show an image of the internal organ, and we are able to see their size and location. In a SPECT scan, one can also see the live function of the target organ. For example, one is able see the pattern of blood flow in the heart. We can also determine which part of the brain is currently active through SPECT. It is done by initially injecting a gamma-emitting radioisotope into your body.

The SPECT scan is similar to Magnetic Resonance Imaging (MRI); however, it is more advanced as it displays live movement. In MRI, we can see a detailed anatomy of the internal organ but not blood flow or the functioning.

Purpose and uses of SPECT scans
SPECT scans are used to display how blood flows into and within tissues and organs. This can help to diagnose and monitor brain, heart or bone-related disorders. This can include seizures, strokes, stress fractures, infections, and tumors in the spine.[7]

I. Neuroimaging or Brain Imaging:

Neuroimaging helps in detecting which part of the brain is affected if you have memory loss, seizure, blood clot in the brain, stroke, or have suffered brain injury or an epileptic attack.

Some specialists also use this imaging technique to detect psychiatric disorders through neuroimaging. For example, it is used for the diagnosis of Attention Deficit Hyperactivity Disorder (ADHD). It is also widely used, either by itself or in combination with MRI, for research purposes. The scanned images serve as data for research with minimal error.

II. Cardiac Imaging:

The imaging technique allows non-invasive imaging of the heart through various methods. SPECT scan can take images of the direction and volume of blood flow. Therefore, it is used to detect any changes in cardiac efficiency. It determines the amount of blood that the heart can pump in a single contraction and the amount that remains in the heart chambers.

It is also used to determine clogged coronary arteries. These are the vessels that supply blood and oxygen to your heart muscles. Sometimes, these vessels develop blockage or become narrow. This can cause permanent damage to the patch of muscles or the muscle fibers. It can be diagnosed early with SPECT scan and then treated.

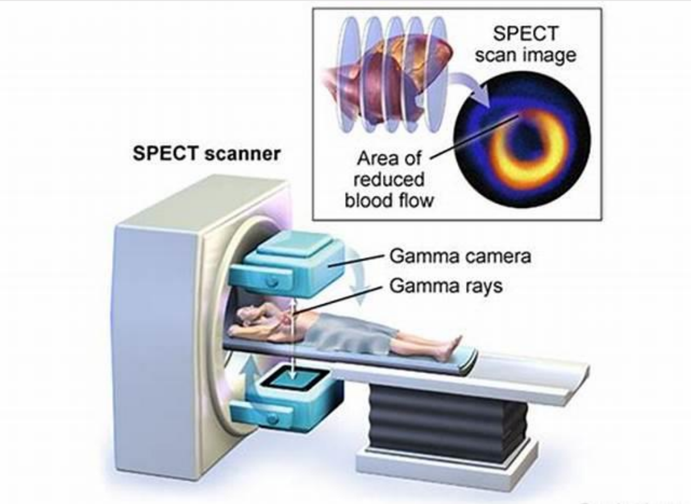

Fig. SP-HM-TRB-2.2: Basic illustration of a SPECT scan of the heart of a patient. (Source: Bing Images)

[7] *(What is a SPECT Scan Commonly Used For?, 2021)*

III. Skeletal Imaging:

SPECT can detect metastasis (progression of cancer) in the bone. It is also used to detect very minute bone fractures that do not show up in a regular X-ray imaging. These fractures are known as hidden fractures. It also shows the area of bone generation or healing. Apart from bone cancer, it can identify small fractures, stress fractures, spinal tumors and bone infections.

2.3. Positron Emission Tomography

Positron imaging utilizes radionuclides that decay by positron emission. The radionuclides used in PET scans are made by attaching a radioactive atom to chemical substances that are used naturally by the particular organ or tissue during its metabolic process. For example, in PET scans of the brain, a radioactive atom is applied to glucose (blood sugar) to create a radionuclide called fluorodeoxyglucose (FDG), because the brain uses glucose for its metabolism. FDG is widely used in PET scanning.

Other substances may be used for PET scanning, depending on the purpose of the scan. If blood flow and perfusion of an organ or tissue is of interest, the radionuclide may be a type of radioactive oxygen, carbon, nitrogen, or gallium.

The radionuclide is administered into a vein through an intravenous line. Next, the PET scanner slowly moves over the part of the body being examined. Positrons are emitted immediately and a very short life time when the radionuclides are introduced and broken down in the patient's body.

When a positron is emitted, it travels less than a millimeter before it collides with an electron in the tissue or organ to be examined. This is referred to as annihilation, which occurs when a particle meets its equivalent antiparticle resulting in them being destroyed and their masses converted to energy. Thus, when the positron and electron annihilate, their mass becomes pure energy in the form of two gamma rays called annihilation photons which move apart in opposite directions and conserving momentum.

The annihilation photons that coincide and travel 180° from each other are then detected by the imaging camera. Similar to SPECT scans, its tomographic images are formed by collecting data from many angles around the patient, resulting in PET images. The amount of the radionuclide collected in the tissue affects how brightly the tissue appears on the image, and indicates the level of organ or tissue function.

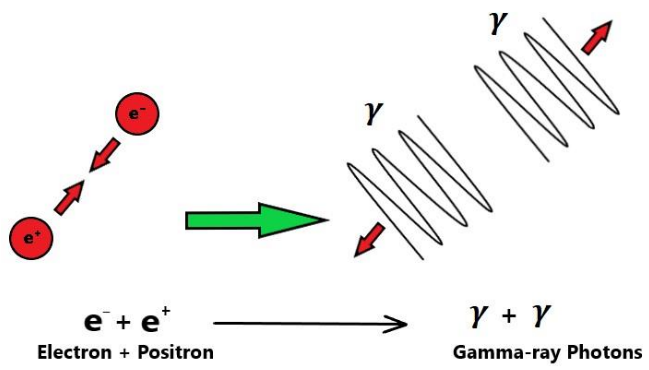

Fig. SP-HM-TRB-2.3: Annihilation of a positron and electron to form two gamma-ray photons (Modified Source: Annihilation in PET Scanning (24.2.2) | CIE A Level Physics Revision Notes 2022 | Save My Exams)

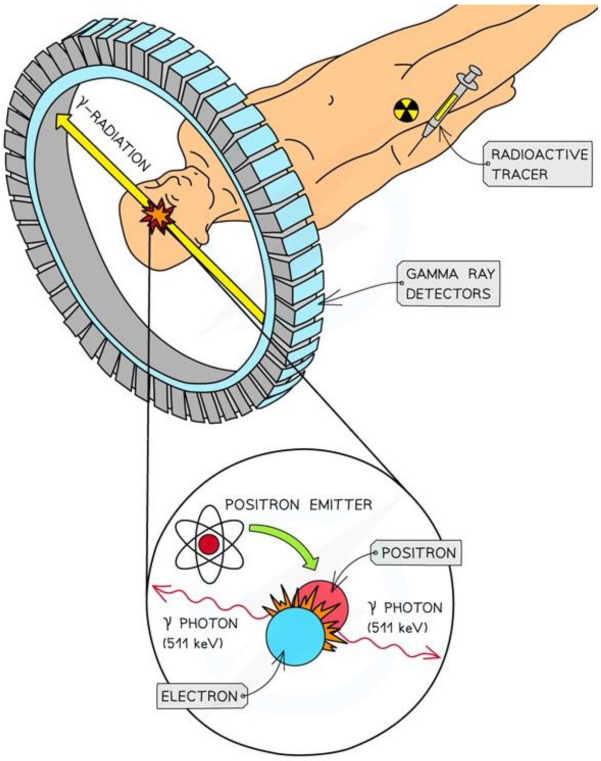

Fig. SP-HM-TRB-2.4: Annihilation of a positron and electron and an electron is the basis of PET scanning. (Modified Source: Annihilation in PET Scanning (24.2.2) | CIE A Level Physics Revision Notes 2022 | Save My Exams)

Purpose and uses of PET Scans

PET scans measure the metabolic activity of the cells within body tissues and this makes it different from other nuclear medicine examinations. PET is actually a combination of nuclear medicine and biochemical analysis. Used mostly in patients with brain or heart conditions and cancer, PET helps to visualize the biochemical changes taking place in the body, such as the metabolism (the process by which cells change food into energy after food is digested and absorbed into the blood) of the heart muscle.

Specifically, PET studies evaluate the metabolism of a particular organ or tissue, so that information about the physiology (functionality) and anatomy (structure) of the organ or tissue is evaluated, as well as its biochemical properties. Thus, PET may detect biochemical changes in an organ or tissue that can identify the onset of a disease process before anatomical changes related to the disease can be seen with other imaging processes such as CT or Magnetic Resonance Imaging (MRI). PET may also be used in conjunction with other diagnostic tests, such as CT or Magnetic Resonance Imaging (MRI). Newer technology combines PET and CT into one scanner, known as PET/CT. PET/CT and provides medical practitioners with a more definitive information about malignant (cancerous) tumors and other lesions *(these are also highlighted in the list below).*

In general, PET scans may be used to evaluate organs and/or tissues for the presence of disease or other conditions. PET may also be used to evaluate the function of organs, such as the heart or brain. The most common use of PET is in the detection of cancer and the evaluation of cancer treatment.

More specific reasons for PET scans include, but are not limited to, the following:

- To diagnose dementias (conditions that involve deterioration of mental function), such as Alzheimer's disease, as well as other neurological conditions such as:
  - Parkinson's disease. A progressive disease of the nervous system in which a fine tremor, muscle weakness, and a peculiar type of gait are seen.
  - Huntington's disease. A hereditary disease of the nervous system which causes increasing dementia, bizarre involuntary movements, and abnormal posture.
  - Epilepsy. A brain disorder involving recurrent seizures.
  - Cerebrovascular accident (stroke).
- To locate the specific surgical site prior to surgical procedures of the brain.
- To evaluate the brain after trauma to detect hematoma (blood clot), bleeding, and/or perfusion (blood and oxygen flow) of the brain tissue.
- To detect the spread of cancer to other parts of the body from the original cancer site
- To evaluate the effectiveness of cancer treatment.
- To evaluate the perfusion (blood flow) to the myocardium (heart muscle) as an aid in determining the usefulness of a therapeutic procedure to improve blood flow to the myocardium.
- To further identify lung lesions or masses detected on chest X-ray and/or chest CT.
- To assist in the management and treatment of lung cancer by staging lesions and following the progress of lesions after treatment.
- To detect recurrence of tumors earlier than with other diagnostic modalities.

Below is a table that outlines some of the differences between PET and SPECT scans.

Table SP-TRB-HM-2.1: Differences between SPECT and PET scans

Positron Emission Tomography (PET)	**Single Photon Emission Computed Tomography (SPECT)**
Uses positron emitting radioisotopes Examples: ^{18}F. ^{68}Ga, ^{64}Cu, ^{18}Zr	Use gamma-ray emitting radioisotopes Examples: ^{99m}Tc, ^{123}I, ^{131}I, ^{123}In
Mainly used for cancer detection and monitoring	Mainly used for heart disease monitoring.
Shorter scanning time than SPECT allowing imaging for multiple areas of body and rapid monitoring of biological processes over short periods of time (e.g., uptake of metabolites into tissues).	Allows the study of slower biological processes (for e.g., uptake of therapeutic antibodies in tissues)
Short half life of some radiotracers allows injection of higher activity leading to increased detection sensitivity	Particles release less energy so radiation dose exposure (dosimetry) can be acceptable.
All radioisotopes have the same energy level so difficult to do simultaneous imaging of different tracers (though methods are being developed)	SPECT isotopes have different energy levels so could permit simultaneous imaging of different processes. This reduces image acquisition times and allows co-registration (correlation) of data in space and time. Contamination between different energy levels is however a concern.
Higher image quality, contrast and spatial resolution compared to SPECT	

High sensitivity (100 – 1000 times higher than SPECT)	
More Costly than SPECT	

Source: Medical imaging: What is the difference between PET and SPECT? - Health Scientific

Activity 1: Review Exercise

Answer the following questions:

Questions

1. **Define the following terms:**
   a. **Emission Tomography**

   ____

   b. **Annihilation electron**

   ____

   c. **Positron**

   ____

2. **What is a SPECT scan? Also include some of its uses and briefly explain what information that these scans provide.**

   ____

3. **What is a PET scan? Also include at least 4 of its uses.**

   ____

4. **Explain what is meant by Annihilation**

5. **What are two differences between a PET and SPECT scan?**

Topic 3: Nuclear Medicine - Therapy

This topic will unravel and explain some of the unique characteristics and clinical applications of therapy treatment using Nuclear Medicine.

Key Points:

- Nuclear medicine therapy, or radionuclide therapy, involves the use of radionuclides to treat serious diseases such as bone metastases and severe cancers.
- The requirements for a therapeutic radionuclide may be divided into two main categories; Physical and biochemical characteristics.
- Physical characteristics include half-life, type of emissions, energy of the radiation(s), daughter product(s), method of production, and radionuclide purity.
- The biochemical aspects include tissue targeting, retention of radioactivity in the tumor, in vivo stability, and toxicity.

3.1. Introduction

Nuclear medicine therapy is an approach to treating cancer that might be used with or after other treatment options, such as chemotherapy and surgery. It will not usually lead to a cure unless combined with other therapies. But for many people it will control symptoms and shrink and stabilize the tumors, sometimes for years. Nuclear medicine therapy is sometimes the best option for people who no longer respond to other treatments.

The use of radioactive molecules as a medication (molecular radiotherapy) that detects malignant cells is what makes nuclear medicine therapy effective. It is administered intravenously, circulates throughout the body, adheres to tumor cells, provides radiation directly to them, and kills them. Some of the drug never binds to cancer cells and floats around in the bloodstream until it is excreted, mostly in the urine. The radioactive drug eventually stops emitting radioactivity and stops killing cancer cells. To get the best results, the processes involved with nuclear medicine therapy is generally done several times.

Nuclear medicine therapy is also called peptide receptor radionuclide therapy (PRRT), targeted radiotherapy or radionuclide therapy.

3.2. Clinical Requirements and Choice of Radionuclides

For decades, scientists have known that radionuclides may be used in therapy. Iodine-131 (^{131}I), Phosphorous-32 (^{32}P), Strontium-90 (^{90}Sr), and Yttrium-90 (^{90}Y) are only a few of the radionuclides that have been effectively employed to treat a variety of cancers. The advent of a variety of new radionuclides and radiopharmaceuticals for the treatment of metastatic bone pain, neuroendocrine, and other malignant or non-malignant tumors has recently fueled the rapid expansion of this sector of nuclear medicine. The field of radionuclide therapy is currently in an exciting phase, with more growth and development expected in the next years. The high frequency of thyroid and liver illnesses in Asia, for example, has generated a slew of new innovations and clinical trials involving tailored radionuclide therapy.

In the last twenty years, radionuclide therapy has been widely used in various clinical malignant and pain management applications. The advantage of radionuclide therapy is that it can give a highly concentrated absorbed dose to the tumor while preserving the surrounding normal tissues. Furthermore, radionuclide therapy's selective ability is beneficial in the treatment of systemic malignancies, such as bone metastases, where whole-body irradiation with external beam radiotherapy is impossible. Targeted radionuclide therapy has become one of the most popular methods of cancer therapy since the administration of radionuclides is minimally invasive and the treatment time is shorter than chemotherapy.

Like all medical drugs, a thorough evaluation is conducted for every radionuclide (or radio-pharmaceutical) before approving its suitability for clinical usage. The requirements for a therapeutic radionuclide may be divided into two main categories, namely physical and biochemical characteristics. Some considerations for physical characteristics include the physical half-life, type of emissions, energy of the radiation(s), daughter product(s), method of production, and radionuclide purity. The biochemical aspects include tissue targeting, retention of radioactivity in the tumor, in vivo stability, and toxicity.

3.2.1. Physical Characteristics

I. Physical Half-life

A suitable range of the physical half-life for therapeutic radionuclides is between 6 hours and 7 days. If it has short physical half-life, it is deemed very impractical as it limits the delivery flexibility. If the half-life is longer than the suitable range, the radiation dose will be retained to the patient and be exposed to surrounding people whom the patient may come in contact with. This means that the patient would need to be admitted and isolated for a longer period, thus increasing their treatment cost.

The biological half-life, on the other hand, is determined by the tracer utilized. The physical half-life of the tracer should not be too long if it is intended to be permanently kept within the patient. For example, the tracer should have sufficient retention so that radiation can be delivered to the tumor efficiently. If the biological half-life is too short, the radionuclide will be discharged with a significantly high activity, resulting in the need for extensive radioactive waste management. Therefore, for an efficient radiation delivery, a balanced optimal biological and physical half-life should be chosen, which results in an optimal effective half-life. This consideration is best made primarily according to the type of tumor to be treated, method of administration, and uptake mechanism. If the uptake is slow, a radionuclide with a relatively long physical half-life should be used so that the radionuclide will not decay completely before it reaches the targeted tumor.

II. Types of Emissions

Radiations with high-linear energy transfer (LET) in therapeutic nuclear medicine, such as α- and β-particles, are preferable. These types of radiations allow very high ionization per length of travel. Therefore, they are fully deposited within a small range of tissue (usually in mm). This also reduces the need for additional radiological protection as the radiation will not penetrate through the patient’s body. However, some β-emitting radionuclides also decay with γ-radiation. This associated γ-radiation could be advantageous if the energy and intensity are within the diagnostic range, as it provides the ability to visualize distribution of the radiopharmaceutical within the patient’s body using gamma scintigraphy methods.

III. Energy of Radiation

Depending on the type of tumor, the energy and intensity of the emitted radiation should be chosen so that the energy and intensity of non-penetrating radiation (i.e., α- and β-particles) are high enough compared with the penetrating radiation (i.e., γ-radiation, X-rays, or Äuger electrons), if present. Radionuclides that emit energetic α- or β-particles are preferred for the treatment of bulky tumors. However, for the treatment of small clusters of cancer cells or small tumor deposits, radionuclides that emit Äuger electrons are considered to be more beneficial because of their high-level cytotoxicity and short-range biological effectiveness.

IV. Daughter of Products

The other factor to be considered is the daughter product of the radionuclide. An ideal radiopharmaceutical should be able to decay into a stable daughter product such as 153-Samarium (^{153}Sm), which fully decays into stable 153 Europium (^{153}Eu). Even if the daughter product is slightly unstable, it should at least be short-lived and may eventually decay within hours into a stable product. However, if the daughter nuclide is not stable, it may contribute to the total amount of absorbed dose. Therefore, we need to consider the percentage of dose delivered from the daughter nuclide so that it will not affect the treatment plan.

3.2.2. Biochemical Characteristics

I. Tissue Targeting and Retention of Radioactivity

For biochemical characteristics, a clinically suitable tracer should provide selective concentrations and prolonged retention in the tumor, while maintaining minimum uptake in the normal tissue. Furthermore, depending on the mechanism of selective uptake of the tumor either by bone deposition, protein binding, or metabolic uptake, the ratio of the radiopharmaceutical's concentration on the tumor to that of other organs or tissues should be as high as possible, so that the radiation can be delivered optimally while minimizing the radiation dose delivered to the normal tissues.

II. In Vivo Stability

If a tracer is a proteinaceous type, it is necessary to ensure that it is stable enough during administration and is maintained in vivo long enough for sufficient radiation delivery within the patient's body before being metabolized and eventually excreted.

The other factor to be considered is the size of the tracer particles. For example, in the radioembolization of liver tumors using 90Y-microspheres, the particles should be large enough so that they will not pass through the capillary bed into other organs, especially the lungs, and at the same small enough so that they will be able to penetrate deep inside the tumor vascularity. Therefore, the typical size for 90Y radioembolization particles is between 20 and 40 μm.

III. Toxicity

Toxicity is another biochemical characteristic that needs to be consider. These considerations include low toxicity, the specific gravity for optimal flow and distribution during administration, the appropriate pH, and the optimal clearance rate (except for permanent tracer).

3.3. Selected Therapeutic Radionuclides and their Clinical Applications

Various radionuclides have been discovered or developed for therapeutic purposes since the first use of radium in the early 1900s. Some examples include Radioactive Iodine Therapy, Bone Metastasis and Radio Immunotherapy.

3.3.1. Radioactive Iodine Therapy

Today, ^{131}I tagged with sodium iodide (^{131}I-NaI) in capsule or liquid form is the most often utilized therapeutic radionuclide. Radioactive Iodine (RAI) therapy is a treatment that uses 131I to treat thyroid-related disorders like Graves' disease, single hyper-functioning nodule, and toxic multinodular goiter. Patients with subclinical hyperthyroidism, especially those at risk of cardiac or systemic problems, may benefit from RAI.

^{131}I has also been used for over 60 years in the treatment of patients with differentiated thyroid cancer. The efficacy of RAI therapy depends on patient preparation, tumor-specific characteristics, the disease site, and the activity of the administered radioiodine. The predictive value for the 10-year relapse-free survival of one negative ^{131}I thyroid scan is about 90% after RAI therapy, whereas two consecutive negative scans have a predictive value greater than 95%. Here, a negative scan means that nothing abnormal was found during the scan.

3.3.2. Treatment of Bone Metastasis

The treatment of bone metastasis is another important application of radionuclide therapy. Several radiopharmaceutical treatments for pain management in bone metastases have been developed in recent decades, with evidence of safety and efficacy *(see Fig. SP-HM-TRB-3.1).* The radiopharmaceutical travels to injured bone volumes only, sparing normal, undamaged bone. The majority of the radionuclides utilized are emitters, such as ^{89}Sr and ^{153}Sm, which release very energetic electrons that deposit their energy in the surrounding tissues over many millimeters. The medication is successful in lowering pain from bone metastases, with pain alleviation reported in 60%–92% of patients in numerous trials. Combining ^{89}Sr with chemotherapy appears to increase pain response and may alter survival in individuals with metastatic prostate cancer, according to several studies.

Radionuclide therapies are also used for the treatment of a variety of malignancies with ^{131}I, ^{90}Y, and ^{188}Re, polycythemia with ^{32}P, chronic joint disease with radio synovectomy, and intravascular radiation therapy.

Table SP-HM-TRB-3.1 summarizes the most commonly used radiopharmaceuticals for targeted radionuclide therapy, including their targeting mechanism and clinical indications.

Table SP-HM-TRB-3.1: Commonly used Radiopharmaceuticals for Targeted Radionuclide Therapy

Radiopharmaceutical	Targeting mechanism	Indication
^{131}I-iodide	Thyroid hormone synthesis	Differentiated thyroid cancer, Graves' disease, hyperfunctioning nodules
^{90}Y-microspheres	Intravascular trapping	Liver metastasis, hepatocellular carcinoma
^{89}Sr-chloride	Calcium analogue	Bone pain palliation
^{153}Sm-EDTMP	Chemo-adsorption	Bone pain palliation

^{90}Y-octreotide	Somatostatin receptor binding	Neuroendocrine tumours
^{131}I-MIBG	Active transport into neuroendocrine cells and intracellular storage	Neuroblastoma, pheochromacytoma, carcinoid, paraganglioma, medullary thyroid carcinoma

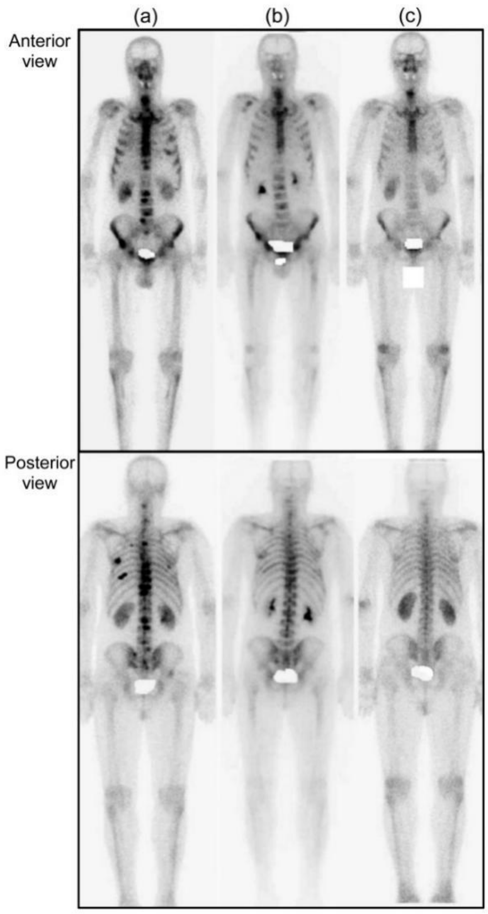

Fig. SP-HM-TRB-3.1: Serial 99mTc methylenediphosphonate (99mTc-MDP) bone scans in a patient with bone metastasis before (a) and at three (b) and six (c) months after combined treatment of 153-Sm oxabifore and denosumab.

From the figure above, we can observe the following:

a) Multiple osteoblastic bone lesions were observed before the treatment.
b) The lesions showed a decreasing trend at the three-month follow-up scan.
c) Complete resolution of the osteoblastic lesions was seen at the six-month follow-up scan

3.3.3. Radio Immunotherapy

As its name suggests, Radio Immunotherapy (RIT) is combination of radiation therapy and immunotherapy. Radiation therapy involves the use of high-energy particles or waves, such as x-rays, gamma rays, electron beams, or protons, to target and destroy cancer cells. Immunotherapy utilizes substances to stimulate or suppress the immune system to help the body fight cancer, infection, and other diseases. RIT utilizes radiopharmaceuticals that contains a combination of a radionuclide and a molecule engineering in a laboratory called a monoclonal antibody (mAbs). These mAbs are able to recognize and bind to specific features of cells such as antigens and cell receptors. In the case for RIT, when the radiopharmaceutical is injected into the patient's bloodstream, it targets tumor-associated antigens and increases the dose delivered, while decreasing the dose to the normal surrounding tissues.

By nature, the technique requires the tumor cells to express an antigen that is unique to the neoplasm - a new and abnormal growth of tissue in a part of the body, especially as a characteristic of cancer - or is inaccessible in normal cells. The concept of RIT was first described by Pressman and Korngold. During the subsequent decades, various RIT radiopharmaceuticals were developed with advances in genetic engineering and chelating techniques.

Over the decades, an increasing number of new antibodies have been studied in clinical trials and have shown positive evidence of efficacy, particularly in non-Hodgkin's lymphoma (NHL). The current US Food and Drug Administration (FDA)-approved radiopharmaceuticals for RIT include ibritumomab tiuxetan and ^{131}I-tositumomab regimen, both for the treatment of refractory NHL. Other potential applications of RIT include the treatment of breast, lung, pancreatic, stomach and ovarian carcinoma, neoplastic meningitis, leukemia, high-grade brain glioma, and metastatic colorectal cancer.

Table SP-HM-TRB-3.2 provides a summary of recently developed mAbs in advanced RIT trials and their status. The clinical outcomes of these radiopharmaceuticals are discussed in the following section.

Table SP-HM-TRB-3.1: Recently developed monoclonal antibodies (mAbs) for advanced RIT

Name	Antibody form	Radionuclide	Antigen	Disease
Ibritumomab tiuxetan (Zevalin)	muIgG1	^{90}Y	CD20	NHL
Tositumomab (Bexxar)	muIgG2 a	^{131}I	CD20	NHL
Epratuzumab (Lymphocide)	huIgG1 (LL2)	^{90}Y	CD22	NHL, CLL, immune diseases
^{131}I-Lym-1 (Oncolym)	huIgG1	^{131}I	HLA-DR10	NHL, CLL
^{131}I-chTNT-1/B (Cotara)	chIgG1	^{131}I	DNA	Glioblastoma multiforme, anaplastic astrocytoma
Labetuzumab (CEA-Cide)	huIgG1	^{90}Y or ^{131}I	CEA	Breast, lung, pancreatic, stomach, and colorectal carcinoma
Pemtumomab (Theragyn)	muIgG1	^{90}Y	PEM	Ovarian, gastric carcinoma
^{131}I-metuximab (Licartin)	Hab18 F(ab')2	^{131}I	Hab18G/CD147	HCC
^{131}I-L19 (Radretumab)	L19	^{131}I	Fibronectin	Hepatological malignancy, refractory Hodgkin's lymphoma, non-small

				cell lung cancer, melanoma, head and neck carcinoma
^{90}Y-clivatuzumab tetraxetan (PAM4)	muIgG1	^{90}Y	MUC1	Pancreatic adenocarcinoma

[*] The information is accurate as at the time of publication
CEA: carcinoembryonic antigen; CLL: chronic lymphocytic leukemia; HCC: hepatocellular carcinoma; HLA: human leukocyte antigen; huIg: human immunoglobulin; muIg: murine immunoglobulin; NHL: non-Hodgkin's lymphoma; PEM: polymorphic epithelial mucin

Discussed below are some of the radionuclides extracted from *Table SP-HM-TRB-3.2*.

I. ^{90}Y-ibritumomab tiuxetan

^{90}Y-ibritumomab tiuxetan (commonly known as Zevalin) is used to treat a certain type of cancer (B-cell non-Hodgkin's lymphoma) in patients whose cancer has returned or has not responded to other treatments. The mAbs in this medication are Ibritumomab tiuxetan and rituximab. They work by killing certain blood and cancer cells from your immune system, particularly the B cells, while the ^{90}Yttrium-90 helps kill the cancer cells.

In 2008, Zevalin was approved for consolidation therapy following partial response or complete response after frontline induction chemotherapy in patients with previously untreated follicular lymphoma in the European Union.

II. ^{131}I-tositumomab

^{131}I-Tositumomabor (commonly known as Bexxar) is an mAb that is labelled with ^{131}I. Bexxar is used to treat certain types of NHL. Investigators have been engaged mainly in defining the role of Bexxar RIT in treating Hodgkin's lymphoma, diffuse large B-cell lymphoma, and multiple myeloma. In a clinical trial of 60 patients with previously untreated follicular lymphoma, Bexxar provided a significant therapeutic efficacy compared with that provided by the last qualifying chemotherapy. Based on the promising outcomes, Bexxar was approved by the FDA for clinical practice in patients with rituximab-relapsed or refractory, low-grade follicular lymphoma in 2003. One year later, the FDA approved Bexxar for RIT in NHL patients without previous rituximab therapy.

III. Epratuzumab

Epratuzumab or Lymphocide is a ^{90}Y radiolabelled mAb directed against CD22 which is an antigen present on the surface of B-cells. Lymphocide may be effective against a variety of diseases that involve abnormal B-cells, including NHL and chronic lymphocytic leukemia. It binds to cells that express the CD22 antigen on their surface and kills the cells by high-energy β^- radiation from the ^{90}Y. Lymphocide is currently being tested in Phase III clinical trials in the treatment of lymphoma, leukaemia, and some immune diseases (e.g., lupus).

Activity 1: Class Discussion Debate

Duration: One Week

In this activity, facilitate a debate by dividing the class into two and allow them to construct compelling arguments around the question provided below:

1. Should PNG adopt and engage Nuclear Medicine use?

Here are two aspects that can be considered during the debate:

- Social Aspect –Health and Wellbeing (Medical)
  - Do the benefits outweigh the possible risks associated with the use of Nuclear Medicine?
  - Does Nuclear medicine pose more of a threat than improve the patient's well-being?
- Economic and Political Aspect
  - What influence to the economy can introducing Nuclear Medicine into medical practices bring into the country?
    - Will there be an increase of people from neighboring countries seeking medical expertise for nuclear medicine use?
  - If the practice of Nuclear Medicine was adopted and introduced as a program for study in the universities in PNG as well as the provision of scholarships to study this abroad, what benefits will this field of medicine provide? Will it contribute **any** significance to the health and welfare in PNG?

Glossary

Antibody	*A blood protein produced in response to and counteracting a specific antigen. Antibodies combine chemically with substances which the body recognizes as alien, such as bacteria, viruses, and foreign substances in the blood.*
Antigen	*A toxin or other foreign substance which induces an immune response in the body, especially the production of antibodies.*
Aquifer	*Body of permeable rock which can contain or transmit groundwater.*
Auxiliary	*Providing supplementary or additional help and support.*
Benign	*(of a disease) not harmful in effect*
Bosons	*A fundamental particle that is responsible for the interactions between elementary particles by carrying and transmitting forces between them.*
Cancer	*A disease caused by an uncontrolled division of abnormal cells in a part of the body.*
Chromatography	*A technique for the separation of a mixture by passing it in solution or suspension through a medium in which the components move at different rates.*
Diagnosis	*the art or act of identifying a disease from its signs and symptoms.*
Electromagnetic Induction	*A current produced because of voltage production (electromotive force) due to a changing magnetic field.*
Electronics	*The branch of physics and technology concerned with the design of circuits using transistors and microchips, and with the behaviour and movement of electrons in a semiconductor, conductor, vacuum, or gas.*
Electrophysiology	*The branch of physiology that deals with the electrical phenomena associated with nervous and other bodily activity.*
Fermions	*Elementary particles that constitute matter.*
Gantry	*Holds radiation detectors and/or a radiation source used to diagnose or treat a patient's illness.*
Genomics	*The branch of molecular biology concerned with the structure, function, evolution, and mapping of genomes.*
Hadron	*a subatomic particle of a type including the baryons and mesons, which can take part in the strong interaction.*
Half-life	
Inductor	*a component in an electric or electronic circuit which possesses inductance.*
Intravenously	*into or by means of a vein or veins*

Isotope	*two or more species of atoms of a chemical element with the same atomic number and nearly identical chemical behavior but with differing atomic mass*
Malignant	*Very virulent or infectious and in the case of tumors. tending to invade normal tissue or to recur after removal; cancerous.*
Mutate	*Change in form or nature.*
Nucleotide	*A compound consisting of a nucleoside linked to a phosphate group. Nucleotides form the basic structural unit of nucleic acids such as DNA.*
Optimum	*Most conducive to a favorable outcome; best.*
Quarks	*Any of a number of subatomic particles carrying a fractional electric charge, postulated as building blocks of the hadrons.*
Radiation	*The emission of energy as electromagnetic waves or as moving subatomic particles, especially high-energy particles which cause ionization.*
Radionuclides	*A distinct kind of atom or nucleus characterized by a specific number of protons and neutrons that emits radiation.*
Physiotherapy	*The treatment of disease, injury, or deformity by physical methods such as massage, heat treatment, and exercise rather than by drugs or surgery.*
Proteomics	*The study of proteomes and their functions.*
Tomography	*A technique for displaying a representation of a cross section through a human body or other solid object using X-rays or ultrasound.*
Transmission	*The mechanism by which power is transmitted from an engine to the axle in a motor vehicle.*
Tumor	*A swelling of a part of the body, generally without inflammation, caused by an abnormal growth of tissue, whether benign or malignant.*

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